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**Statistical Forecasting of Exchange Rates: A
Review**
**A Comparison of Standard Time Series Model and
Machine Learning Technique**

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ABSTRACT

Forecasting exchange rates is a vital component of the global financial system, influencing international trade, market investments, and monetary policy.

This thesis investigates the performance of traditional time series models and machine learning approaches in the predicting of features of exchange rate fluctuations, such as, price

level, returns, direction, etc. (think about other different features that you could be interested in comparing) with an emphasis on identifying the contexts in which each methodology is most effective. The study will test formally the accuracy of machine learning techniques and time series models at capturing patterns within exchange rate data, in terms of various evaluation metrics. The results of this research contribute to the understanding of how different forecasting methods perform under varying conditions. Insights gained are intended to aid practitioners in finance, economics, and policymakers in the selection of the most suitable tools for enhancing prediction accuracy, and thereby supporting better strategic decisions.

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INTRODUCTION

1. Literature review

1.1. Exchange Rate Dynamics and Their Global Significance

Exchange rates play a fundamental role in shaping the global economy, influencing trade, investments, and financial stability. They represent the relative value of one currency against another and are affected by a range of macroeconomic and geopolitical factors. Fluctuations in exchange rates can impact a country's economic performance, affecting inflation, trade balances, and capital flows. Understanding these dynamics is essential for businesses, policymakers, and investors navigating the complexities of international finance (Krugman & Obstfeld, 2014).

Exchange rate fluctuations are driven by several key factors, including economic indicators, market sentiment, and policy decisions. Macroeconomic fundamentals: indicators such as GDP growth, inflation, and employment levels influence currency valuation. Countries with strong economic performance typically see their currencies appreciate, while weaker economies may experience depreciation (Mankiw, 2018). Interest rate differentials: higher interest rates tend to attract foreign investment, increasing demand for a currency and strengthening its value. Conversely, lower interest rates can lead to capital outflows and depreciation (Dornbusch, Fischer & Startz, 2018). Central bank policies: monetary policy decisions, such as adjusting interest rates or implementing quantitative easing, can significantly impact exchange rates. Central banks may also intervene in foreign exchange markets to stabilize their currencies (Mishkin, 2019). Market speculation and investor Sentiment: short-term fluctuations often result from speculative trading and investor expectations about future economic conditions (Frankel & Rose, 1995). Trade balances: a country with a trade surplus—exporting more than it imports—typically experiences currency appreciation. In contrast, trade deficits can lead to depreciation as more currency flows out of the country (Krugman & Obstfeld, 2014). Political and economic stability: Political uncertainty, conflicts, and economic crises can trigger sharp currency devaluations, as investors seek safer assets (Reinhart & Rogoff, 2009).

Exchange rates influence various aspects of international trade and investment. A depreciating currency can make a country's exports more competitive but

may increase the cost of imported goods, potentially fueling inflation. Conversely, an appreciating currency makes imports cheaper but can hurt export competitiveness (Krugman & Obstfeld, 2014).

Foreign direct investment (FDI) is also affected by exchange rate stability. Investors prefer economies with predictable exchange rate movements, as excessive volatility can pose financial risks (Mankiw, 2018). Additionally, currency fluctuations influence financial markets, impacting stock prices, commodity values, and global capital flows (Dornbusch et al., 2018).

Countries adopt different exchange rate regimes based on their economic policies and priorities. Fixed exchange rate systems: A currency is pegged to another, such as the U.S. dollar, ensuring stability but limiting monetary policy flexibility (Frankel & Rose, 1995). Floating exchange rate systems: Market forces determine exchange rates, offering flexibility but increasing volatility (Krugman & Obstfeld, 2014). Managed float (hybrid systems): Central banks intervene when necessary to stabilize excessive fluctuations, balancing flexibility and stability (Reinhart & Rogoff, 2009).

1.2. Traditional vs. Machine Learning Approaches in Forecasting

Forecasting is an essential tool across various domains, including finance, economics, weather prediction, and supply chain management. Over time, two primary approaches have emerged: traditional statistical models and modern machine learning techniques. While both aim to generate accurate predictions, they differ significantly in their methodology, underlying assumptions, and effectiveness in capturing complex patterns.

Traditional forecasting relies on well-established statistical models that assume a structured relationship between past and future values. These models are widely used due to their mathematical foundation and interpretability. Some of the most common methods include: Autoregressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), Exponential Smoothing (ETS), Nonlinear Autoregressive Distributed Lag (NARDL)

Autoregressive Integrated Moving Average (ARIMA): A time series model that accounts for trends, seasonality, and noise by combining autoregression, differencing, and moving averages (Box & Jenkins, 1970).

Vector Autoregression (VAR): A multivariate model that captures interdependencies between multiple time series, often applied in macroeconomic forecasting (Stock & Watson).

Exponential Smoothing (ETS): A technique that assigns decreasing weights to older observations, making it particularly effective for capturing trends and seasonal effects (Hyndman & Athanasopoulos, 2018).

Nonlinear Autoregressive Distributed Lag (NARDL): Designed to model asymmetric relationships in time series data, making it useful for financial and economic analysis (Shin et al., 2014).

Despite their widespread use, these models have limitations. Many assume linear relationships, stationarity, or predefined structures, which may not always hold in real-world data. Their ability to adapt to changing patterns is also constrained, requiring careful specification and parameter tuning.

In contrast, machine learning (ML) models offer a more flexible approach by automatically detecting complex, nonlinear patterns in data. These methods do not rely on strict assumptions, making them particularly effective for dynamic and high-dimensional datasets. In these research work we are considering some of the ML techniques.

Long Short-Term Memory (LSTM) Networks: A type of recurrent neural network (RNN) designed to capture long-term dependencies in sequential data, making it highly effective for time series forecasting (Chollet, 2018).

Gradient Boosting Algorithms (e.g., XGBoost, LightGBM): these ensemble learning methods build multiple decision trees sequentially, refining errors at each step to improve prediction accuracy (Chen & Guestrin, 2016).

Convolutional Neural Networks (CNNs) for Time Series: Originally developed for image recognition, CNNs have been adapted for time series forecasting by identifying temporal patterns (Borovykh et al., 2017).

Variational Mode Decomposition (VMD) Hybrid Models: A signal-processing technique that decomposes time series into intrinsic components, improving predictive performance when combined with ML models (Dragomiretskiy & Zosso, 2014).

ML-based forecasting methods excel at capturing nonlinear relationships and handling large volumes of data. However, they require substantial computational resources, large datasets, and careful hyperparameter tuning. Additionally, many ML models function as "black boxes," making it difficult to interpret their predictions—a critical drawback in domains like finance, where transparency is essential.

1.3. Key Challenges in Exchange Rate Prediction

Forecasting exchange rates is a complex and demanding task due to the dynamic nature of currency markets. Various economic, political, and external factors contribute to sudden fluctuations, making it difficult to develop consistently accurate predictive models. The main challenges in exchange rate forecasting stem from high volatility, structural breaks, and external shocks, all of which can significantly impact currency movements.

One of the most prominent challenges in exchange rate forecasting is the high level of volatility in currency markets. Exchange rates are influenced by a range of factors, including interest rates, inflation, geopolitical developments, and speculative trading. Unlike stock markets, where prices follow an opening and closing pattern, currency markets operate 24/7, leading to continuous fluctuations in value.

The unpredictable nature of these fluctuations poses difficulties for traditional forecasting models, particularly those relying on linear assumptions. For instance, standard time series models like ARIMA struggle to capture the nonlinear and chaotic behavior of exchange rates. To address this, more sophisticated approaches, such as Generalized Autoregressive Conditional Heteroskedasticity (GARCH) (Engle, R. F., 1982) models, have been introduced to model volatility patterns. However, even these methods may fail to predict sudden price swings triggered by unexpected market developments.

Another major obstacle in exchange rate prediction is the occurrence of structural breaks — sudden shifts in the underlying economic environment that disrupt previously established trends. These breaks can be caused by various factors, including changes in monetary policy, financial crises, or major geopolitical events. When such shifts occur, traditional forecasting models

often become unreliable, as they assume that past patterns will continue into the future.

For example, during the 2008 Global Financial Crisis, exchange rates exhibited drastic and unpredictable movements, rendering many pre-existing models ineffective. To account for these disruptions, researchers have developed Markov switching models (Hamilton, J. D., 1989), which allow for different economic regimes in forecasting. Similarly, Bayesian change point detection (Barry, D., & Hartigan, J. A., 1993) helps identify structural breaks and adjust predictions accordingly. Despite these advancements, accurately forecasting the timing and magnitude of such shifts remains a significant challenge.

Exchange rates are also highly sensitive to external shocks—unpredictable events that can cause sharp currency fluctuations. These shocks can arise from various sources, including global financial instability, commodity price swings, and natural disasters. For instance, the COVID-19 pandemic led to extreme volatility in currency markets due to disruptions in trade and capital flows.

Countries reliant on commodity exports are particularly vulnerable to price fluctuations in global markets. A sudden drop in oil prices, for example, can lead to a sharp depreciation in the currency of an oil-exporting nation. Similarly, political instability, such as Brexit or international sanctions, can create uncertainty that affects investor confidence and exchange rate stability.

While some forecasting models attempt to incorporate real-time macroeconomic indicators, the unpredictable nature of these external shocks makes accurate forecasting difficult. In recent years, event-driven models and news sentiment analysis have been explored as potential methods for capturing real-time market reactions, but their effectiveness is still under evaluation.

1.4. ARIMA and Its Applications

The Autoregressive Integrated Moving Average (ARIMA) model is a widely used statistical technique for time series forecasting, particularly effective for datasets that exhibit stationarity or can be transformed into a stationary form. Its versatility has made it a fundamental tool in economic and financial forecasting, including exchange rate prediction.

ARIMA models are defined by three key parameters:

- **p (autoregressive component):** Represents the influence of past values on the current observation.
- **d (degree of differencing):** Specifies the number of times the series must be differenced to achieve stationarity.
- **q (moving average component):** Captures the relationship between past forecast errors and the current value.

By adjusting these parameters, ARIMA models can be tailored to fit various types of time series data, enabling accurate short-term forecasting.

Developed by Box and Jenkins (1970), this systematic approach provides a structured framework for building ARIMA models. It consists of three primary steps:

1. Model identification

- 1) Determine whether the data is stationary or requires differencing.
- 2) Use autocorrelation function (ACF) (Dickey, D. A., & Fuller, W. A., 1979) and Partial Autocorrelation Function (PACF) (Hyndman, R. J., & Athanasopoulos, G., 2018) plots to select appropriate values for p and q.

2. Parameter estimation: estimate model coefficients using Maximum Likelihood Estimation (MLE) (Ronald A. Fisher, 1921) or Least Squares (Adrien-Marie Legendre, 1805) methods.

3. Model Validation

- 1) Analyze residuals to ensure they resemble white noise (random, uncorrelated errors).
- 2) Compare different models using selection criteria like Akaike Information Criterion (AIC) (Akaike, H., 1974) or Bayesian Information Criterion (BIC) (Gideon E. Schwarz, 1978) to identify the most suitable one.

This methodology provides a systematic way to construct reliable ARIMA models while ensuring statistical rigor in forecasting.

A critical assumption of ARIMA is that the time series must be stationary, meaning its statistical properties (mean, variance, and autocovariance) remain constant over time. If a series is non-stationary, differencing (d) is applied to remove trends and seasonality.

Indicators of non-stationarity include: trends (long-term upward or downward movements in the data); seasonal Effects (repeating patterns at regular intervals); presence of a Unit Root (identified using tests such as the Augmented Dickey-Fuller (ADF) test (1979) or Phillips-Perron (PP) test (1988)).

In cases where differencing alone is insufficient, logarithmic transformations or the Box-Cox transformation can help stabilize variance and improve model performance.

In a seminal study, Meese and Rogoff (1983) compared ARIMA with fundamental economic models, such as purchasing power parity (PPP) and interest rate differentials. Their findings revealed that ARIMA often provided better short-term forecasts, yet lacked explanatory power beyond simple data-driven predictions.

Research by Zhang and Hu (1998) highlighted ARIMA's limitations when handling nonlinear relationships in financial data. Their study compared ARIMA with neural networks, showing that while ARIMA excelled in short-term linear trend forecasting, machine learning models were better suited for capturing complex, nonlinear dynamics.

Chong, Ahmad, and Abdullah (2014) analyzed ARIMA's forecasting accuracy in emerging market currencies. Their study found that while ARIMA could produce reliable short-term forecasts, frequent re-estimation was necessary due to high market volatility and structural shifts, reducing its long-term effectiveness.

Hyndman and Athanasopoulos (2018) emphasized ARIMA's reliance on stationarity and linearity, noting that it may not always be the best choice for highly volatile financial markets. They suggested combining ARIMA with alternative models, such as exponential smoothing or hybrid machine learning approaches, for improved accuracy.

1.5. The Rise of Machine Learning in Financial Forecasting

In recent years, the application of machine learning (ML) to financial forecasting has gained significant attention due to its potential to improve accuracy and capture complex patterns in financial markets that traditional models might miss. Machine learning techniques, particularly deep learning models such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), have shown great promise in addressing some of the challenges posed by time series forecasting in finance, including volatility, non-linearity, and high-dimensional data (Bengio et al., 2013; LeCun et al., 2015). This section will explore the rise of machine learning in financial forecasting, focusing on recent advancements in the field, as well as the integration of these methods with traditional statistical approaches and the key advantages of machine learning when dealing with non-linear and high-dimensional data, emphasizing its flexibility, scalability, and ability to uncover hidden structures.

Financial forecasting, especially for exchange rates, stock prices, and other market indicators, is a critical task in economic and financial decision-making. Traditionally, statistical models such as AutoRegressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), and Exponential Smoothing (ETS) were extensively used to model financial time series data (Hyde, 2012). These models are based on assumptions of linearity, stationarity, and the existence of trends and seasonality, which may not always hold in real-world financial data.

In contrast, machine learning methods offer more flexibility, capable of modeling complex, non-linear relationships within the data. These techniques, such as Support Vector Machines (SVM), Random Forests, Gradient Boosting, and Neural Networks, can adapt to the changing patterns in financial markets by learning from large amounts of data. Moreover, ML models can handle high-dimensional data and make use of unstructured data such as news articles, social media sentiment, and financial reports (Chen et al., 2018).

Recent advancements in deep learning, particularly in the field of Recurrent Neural Networks (RNNs), have revolutionized financial forecasting. Among these, Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) and Gated Recurrent Units (GRUs) (Cho et al., 2014) have emerged as the dominant architectures for time series prediction. LSTM and GRU models are designed to overcome the limitations of traditional RNNs by

effectively handling long-range dependencies, which are often present in financial time series data.

LSTMs and GRUs work by maintaining a memory of past time steps and selectively "forgetting" irrelevant information. This ability makes them particularly suited for modeling the temporal dependencies in financial data, such as the relationships between past exchange rates and future trends. Recent studies have demonstrated the superior performance of LSTM and GRU models over traditional time series models (Krauss et al., 2017; Fischer & Krauss, 2018), especially in volatile and noisy environments typical of financial markets.

While deep learning models such as LSTM and GRU are powerful tools for financial forecasting, they can also be prone to overfitting, particularly in cases of limited data. Consequently, researchers have explored hybrid models that combine machine learning algorithms with traditional statistical methods to improve forecasting performance. For example, some studies have proposed hybrid ARIMA-LSTM models (Zhang et al., 2020), which leverage the strengths of both linear modeling and deep learning for exchange rate prediction. These hybrid models have been shown to outperform standalone models in terms of both accuracy and robustness.

Another challenge in financial forecasting is model interpretability. While machine learning models, particularly deep learning techniques, tend to be black-box models, recent developments have focused on improving their interpretability. Methods such as SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) and Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016) have been applied to financial forecasting to provide insights into model decision-making, enabling financial analysts and decision-makers to trust and better understand the predictions.

The application of machine learning extends beyond time series forecasting in financial markets. One notable area of research is sentiment analysis, which involves analyzing news articles, social media, and other textual data to gauge market sentiment. This technique has gained significant traction in the context of stock market prediction, where sentiment analysis models are used to predict price movements based on the sentiment conveyed in financial news or tweets (Zhang et al., 2018). Similarly, anomaly detection and fraud detection systems

built on machine learning have become essential in ensuring the integrity of financial transactions (Chandola et al., 2009).

Moreover, the development of new forecasting techniques such as the Variational Mode Decomposition (VMD) model (Dragomiretskiy & Zosso, 2014) combined with deep learning networks has added an additional layer of sophistication to the forecasting process. The VMD method, which decomposes a time series into intrinsic mode functions, can enhance the accuracy of machine learning models, especially when forecasting volatile and noisy data such as exchange rates (Mitra et al., 2021).

Despite the promising advancements in machine learning for financial forecasting, several challenges remain. One of the primary challenges is the quality and availability of data. Financial markets are subject to frequent disruptions, and data can be noisy, non-stationary, and subject to regime shifts. Additionally, the lack of labeled data for supervised learning methods remains an obstacle for training effective machine learning models.

Furthermore, deep learning models require significant computational resources, which may be a limitation for smaller financial institutions or individual traders. There is also the risk of overfitting when working with complex models and insufficient data.

To address these challenges, future research is likely to focus on improving data quality through advanced data preprocessing techniques, developing more efficient algorithms, and creating models that can adapt to changing market conditions in real-time. Additionally, there is growing interest in the integration of reinforcement learning (RL) with machine learning for predictive modeling and decision-making in financial markets (Deng et al., 2016).

Machine learning (ML) has revolutionized data analysis and predictive modeling by offering powerful tools to handle complex, non-linear, and high-dimensional data. Traditional statistical models often struggle with such data due to assumptions of linearity and stationarity, whereas ML methods are designed to adaptively learn patterns without requiring strict predefined relationships.

One of the most significant advantages of ML models is their ability to capture non-linear relationships between variables. Traditional statistical models such

as linear regression or ARIMA assume linear dependencies, which may not accurately reflect real-world systems. Many phenomena, especially in finance, climate science, and healthcare, involve intricate dependencies that cannot be adequately modeled using linear equations (Hastie et al., 2009).

Neural networks, support vector machines (SVMs), and tree-based methods such as Random Forests and Gradient Boosting Machines (GBMs) excel at capturing these complex relationships. For instance, deep learning models, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, can automatically learn intricate patterns in data without requiring manual feature engineering (LeCun et al., 2015). These models utilize activation functions such as ReLU, sigmoid, and tanh, allowing them to represent highly non-linear functions effectively (Goodfellow et al., 2016).

High-dimensional data refers to datasets with a large number of features (variables). As the number of dimensions increases, traditional statistical methods often suffer from the "curse of dimensionality," which can lead to overfitting, increased computational complexity, and poor generalization (Bellman, 1957). Machine learning techniques, particularly deep learning and dimensionality reduction methods, mitigate these issues through feature selection, regularization, and automated feature extraction.

Machine learning offers several powerful dimensionality reduction techniques, including:

- Principal Component Analysis (PCA): Reduces data dimensionality by transforming features into principal components that retain the most variance (Jolliffe & Cadima, 2016).
- Autoencoders: Neural networks that learn compact representations of high-dimensional data by encoding and reconstructing input data (Hinton & Salakhutdinov, 2006).
- t-Distributed Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP): Non-linear techniques that preserve local and global structure in high-dimensional data (Van der Maaten & Hinton, 2008).

These techniques allow ML models to work efficiently with high-dimensional datasets while preserving meaningful information.

Machine learning models are particularly robust when working with large, unstructured, and noisy datasets. Unlike traditional models that require strict assumptions about data distributions, ML methods can learn directly from raw data, filtering out noise and identifying meaningful patterns.

For example, deep learning models like CNNs have been successfully used in image recognition tasks involving millions of pixels per image (Krizhevsky et al., 2012). Similarly, ensemble methods such as Random Forests and XGBoost reduce the impact of noisy features by averaging multiple models, leading to more stable predictions (Chen & Guestrin, 2016).

One of the core strengths of ML models is their ability to automatically extract and learn meaningful representations from raw data. Traditional statistical models require manual feature selection, which can be time-consuming and prone to bias. In contrast, deep learning models, particularly CNNs and LSTMs, can automatically learn hierarchical representations without explicit feature engineering.

For instance:

1. CNNs extract spatial features in images (LeCun et al., 1998).
2. LSTMs and GRUs process sequential data, capturing long-term dependencies (Hochreiter & Schmidhuber, 1997).
3. Transformer-based models, such as BERT and GPT, extract semantic meaning from textual data (Vaswani et al., 2017).

This ability to learn features autonomously makes ML models highly effective across diverse applications.

High-dimensional data often leads to overfitting, where a model performs well on training data but poorly on new, unseen data. Machine learning models incorporate various regularization techniques to prevent overfitting and improve generalization, such as:

- L1 and L2 Regularization (ridge and lasso regression): Reduce model complexity by penalizing large coefficients (Tibshirani, 1996).

- Dropout in neural networks: prevents over-reliance on specific neurons by randomly deactivating nodes during training (Srivastava et al., 2014).
- Batch Normalization: Stabilizes learning by normalizing activations across layers (Ioffe & Szegedy, 2015).

These techniques help ML models make accurate predictions on new data.

Machine learning algorithms, particularly deep learning models, can scale efficiently with large datasets. Unlike traditional econometric models, which may struggle with high-dimensional data, ML models leverage parallel computing and GPUs for fast training and inference (Goodfellow et al., 2016).

For instance, cloud-based ML platforms such as Google TensorFlow and PyTorch enable distributed computing, making it feasible to train complex models on massive datasets (Abadi et al., 2016). This scalability is crucial for applications in finance, healthcare, and social media, where real-time processing of large data streams is necessary.

Machine learning models are highly versatile and have been successfully applied to a wide range of fields, including:

1. Finance: predicting stock prices, exchange rates, and credit risk assessment (Zhang et al., 2020).
2. Healthcare: diagnosing diseases and predicting patient outcomes using medical imaging and electronic health records (Esteva et al., 2017).
3. Natural language processing (NLP): sentiment analysis, machine translation, and chatbots (Devlin et al., 2019).
4. Autonomous systems: self-driving cars and robotics (Bojarski et al., 2016).

The ability of ML models to generalize across different domains underscores their robustness in handling complex, high-dimensional data.

2. Methodology

2.1. Exploratory Data Analysis

2.1.1. Linear Interpolation

Handling missing values is a crucial step in time series analysis, as incomplete data can lead to biased results and reduced forecasting accuracy. Several imputation techniques exist, ranging from simple statistical methods to complex machine learning-based approaches. Among these, linear interpolation is a widely used technique due to its simplicity, computational efficiency, and ability to maintain temporal consistency in time series data.

Linear interpolation is chosen as an imputation method because it ensures a smooth transition between missing values while preserving the overall trend of the data. Unlike mean or median imputation, which replace missing values with a constant, linear interpolation takes into account the temporal ordering of observations. This makes it particularly suitable for time-dependent data, where sequential relationships play a critical role (Hyndman & Athanasopoulos, 2018).

Other methods, such as listwise deletion, may lead to significant data loss, reducing the statistical power of models. More advanced techniques, such as regression imputation or machine learning-based methods, can introduce unnecessary complexity without significant gains in accuracy when missing values are limited (Jadhav et al., 2019). Linear interpolation strikes a balance between accuracy and simplicity, making it a practical choice for time series preprocessing.

Linear interpolation estimates a missing value X_t at time t based on its two nearest values, X_{t-1} and X_{t+1} , using the following formula:

$$X_t = X_{t-1} + \frac{X_{t+1} - X_{t-1}}{(t+1) - (t-1)} \times (t - (t-1)) \quad (1)$$

where:

X_t is the missing value,

X_{t-1} and X_{t+1} are the nearest known values before and after t ,
 t represents the time index.

This method assumes that changes between consecutive data points occur in a linear manner, which is a reasonable assumption for many financial and economic time series that exhibit smooth trends over short periods (Taylor, 1986).

Linear interpolation provides a simple yet effective solution for handling missing values in time series data. It avoids excessive data loss, maintains the integrity of temporal relationships, and ensures smooth transitions between observed values. Compared to alternative imputation techniques, it offers a practical balance between computational efficiency and accuracy, making it a preferred choice in many forecasting applications.

2.1.2. Log Returns and Squared Log Returns in Time Series Analysis

In the context of exchange rate forecasting, the use of log returns instead of raw prices is standard practice in empirical finance and econometrics. Logarithmic returns, also known as continuously compounded returns, are calculated as the natural logarithm of the ratio of successive prices.

Log returns, also known as continuously compounded returns, offer an alternative approach that addresses the limitations of simple returns. They are particularly useful in time series modeling because they are additive over time, allowing for more efficient calculations in forecasting models (Cont, 2001). The formula for log returns is:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1} \quad (2)$$

Where:

r_t is the log return at time t

P_t and P_{t-1} are the exchange rates at time t and $t - 1$,

$\ln$ denotes the natural logarithm.

Log returns are advantageous for several reasons. They stabilize variance in financial time series, making the data more suitable for statistical analysis. They also approximate normality under certain conditions, which benefits many

econometric and machine learning models (Fama, 1965). Additionally, log returns enable more accurate performance evaluation over long time horizons, as they naturally incorporate compounding effects.

The decision to use simple or log returns depends on the objectives of the analysis:

- Simple returns are easy to interpret and widely used in finance when assessing short-term price changes.
- Log returns are preferred in forecasting models due to their mathematical properties, such as additivity and improved stationarity.

Since financial time series models, including ARIMA and GARCH, require stationary data, log returns are often the preferred choice for forecasting applications (Tsay, 2010).

To capture volatility and risk patterns in the exchange rate movements, squared log returns are also considered. Squaring the log return amplifies larger deviations and neutralizes the direction of the return, making it a useful proxy for volatility. The squared return r_t^2 highlights periods of market turbulence and is frequently used in models such as GARCH to estimate time-varying variance.

2.1.3. Simple Moving Average (SMA) for Trend Analysis

Exchange rate movements are often characterized by short-term fluctuations, which can obscure the detection of long-term trends. To address this, the Simple Moving Average (SMA) is applied as a smoothing technique to filter out noise and reveal the underlying direction of the exchange rate. The SMA is a widely used statistical method in time series analysis, offering a straightforward yet effective way to identify trends.

The SMA at a given time t is calculated as the arithmetic mean of the previous n observations, defines as:

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} X_{t-i} \quad (3)$$

where X_t represents the exchange rate at time t , and n denotes the length of the averaging window. In this study, a 30-day window is chosen to ensure that short-term volatility does not obscure broader market trends.

The SMA is particularly useful in exchange rate forecasting as it helps identify persistent trends by mitigating the impact of temporary fluctuations. Unlike more sophisticated filtering techniques, such as the Exponential Moving Average (EMA) or autoregressive models, the SMA assigns equal weight to all values within the window. This characteristic makes it simple to interpret while still providing meaningful insights into trend direction (Hyndman & Athanasopoulos, 2018).

However, a notable limitation of SMA is its inherent lag, which can delay the recognition of turning points in the exchange rate trajectory. Since all values within the selected window contribute equally to the average, SMA responds slowly to sudden market shifts, potentially limiting its effectiveness in rapidly changing economic conditions (Shumway & Stoffer, 2017).

The primary advantage of SMA lies in its ability to smooth short-term price variations, making it easier to observe general trends in the foreign exchange market. Additionally, its computational simplicity makes it a practical tool for both financial analysts and researchers. Furthermore, SMA serves as a benchmark for comparison with more advanced time series forecasting models, such as Autoregressive Integrated Moving Average (ARIMA) and machine learning-based approaches (Chatfield, 2003).

Despite its usefulness, SMA has drawbacks. Since it gives equal importance to all past values in the window, it may not adapt quickly to sudden changes in exchange rates. This limitation makes it less effective during periods of high volatility when quick adjustments are necessary for accurate forecasting. In contrast, models like the EMA, which place greater emphasis on recent observations, may offer improved responsiveness in such scenarios.

By employing the SMA as an initial analytical tool, this study establishes a baseline for understanding exchange rate dynamics before incorporating more complex predictive models.

2.1.4. Classical and STL (Seasonal-Trend using Loess) Decompositions

This section describes the analytical framework for decomposing time series data using two key methods: Classical Decomposition and STL (Seasonal-Trend decomposition using Loess). These techniques enable the separation of a time series into interpretable elements, supporting a deeper understanding of the structural patterns embedded in the data.

Time series data often contain a combination of long-term movement, periodic fluctuations, and irregular noise. Decomposition methods aim to isolate these components to better understand and model the data. The series is typically split into:

Trend: the long-run progression or direction of the data.

Seasonal component: systematic, calendar-based variation.

Residual (or irregular) component: unexplained noise or randomness.

Both Classical and STL decomposition techniques rely on this conceptual structure but differ significantly in terms of flexibility, assumptions, and estimation methods.

The classical approach assumes that the time series can be expressed in either an additive or multiplicative form, depending on whether seasonal variations remain constant or scale with the level of the series. The two models are structured as follows:

Additive:

$$y_t = T_t + S_t + R_t \quad (4)$$

Multiplicative:

$$y_t = T_t \cdot S_t \cdot R_t \quad (5)$$

Here, y_t is the observed value, T_t the trend, S_t the seasonal component, and R_t the residual at time t . In practice, the trend is typically estimated using moving averages. Once the trend is removed, seasonal effects are derived by averaging the detrended values across corresponding seasonal positions (e.g.,

months or quarters). Finally, residuals are obtained by subtracting (or dividing) the estimated trend and seasonality from the original series.

While classical decomposition is easy to implement and interpret, it assumes a fixed seasonal structure over time, which may not reflect real-world dynamics in more volatile or evolving datasets (Makridakis et al., 1998).

STL, introduced by Cleveland et al. (1990), offers a more adaptable and robust alternative. It decomposes the series into trend, seasonal, and residual components using Loess smoothing, a form of locally weighted regression. The key feature of STL is its ability to handle non-static seasonal patterns, making it particularly useful when seasonal effects change over time. The decomposition follows the additive form (8).

STL operates through an iterative process that alternately estimates the seasonal and trend components. It allows users to control the degree of smoothing applied to both, enhancing customization. Furthermore, STL can be made robust to outliers, improving its reliability in the presence of noise or structural breaks.

Unlike classical decomposition, STL does not require pre-specification of seasonal strength and can adapt to changing data behavior, which is particularly valuable in financial and economic contexts where conditions shift frequently.

Both decomposition methods serve as foundational tools for exploratory time series analysis. Classical decomposition is suitable when the seasonal pattern is known to be stable and the focus is on interpretability. STL is better suited to more complex series with evolving structures and varying seasonality.

By examining the trend and seasonal components extracted from each method, researchers can identify periods of structural change, detect anomalies, and inform the selection of appropriate forecasting or modeling techniques.

2.2. ARIMA – Autoregressive Integrated Moving Average

The Autoregressive Integrated Moving Average (ARIMA) model is one of the most widely used methods for time series forecasting. Introduced by Box and Jenkins (1970), ARIMA combines three components: autoregression (AR), differencing (I) to achieve stationarity, and a moving average (MA) process to

capture noise dependencies. Its ability to model both stationary and non-stationary data has made it a fundamental tool in economics, finance, and business forecasting.

This review examines its theoretical framework, applications in financial forecasting, and comparisons with alternative models.

The ARIMA model, expressed as ARIMA(p, d, q), consists of: p - the number of autoregressive (AR) terms, reflecting past values' influence; d - the degree of differencing, making a time series stationary; q - the number of moving average (MA) terms, capturing error dependencies.

Mathematically, an ARIMA process is represented as:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t \quad (6)$$

Where y_t is the time series at time t, ϕ_i are AR coefficients, θ_j are MA coefficients and ϵ_t is a white noise error term.

The Box-Jenkins methodology outlines a systematic approach for identifying and estimating ARIMA models: identification is used of Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to determine optimal p and q. Estimation: Parameters are computed via Maximum Likelihood Estimation (MLE) or Least Squares Estimation. Diagnostics: residuals must be uncorrelated and normally distributed for model validity.

In *Forecasting: Principles and Practice* (3rd ed.), Hyndman & Athanasopoulos (2021) provide a detailed overview of ARIMA, emphasizing its role in modeling stationary and non-stationary time series.

A fundamental requirement for ARIMA is stationarity, meaning that statistical properties such as mean and variance remain constant over time. When a series exhibits trends or seasonality, differencing (the "I" component) is applied to stabilize fluctuations. The authors recommend the Augmented Dickey-Fuller (ADF) test to determine whether differencing is necessary, ensuring the model does not overfit or misinterpret trends.

The effectiveness of an ARIMA model depends on the correct choice of autoregressive (p) and moving average (q) terms. Hyndman & Athanasopoulos (2021) advocate for analyzing Autocorrelation Function (ACF) and Partial

Autocorrelation Function (PACF) plots, which provide insights into how past values influence the series. Overfitting remains a concern, as excessive lag selection can reduce the model's generalizability.

While ARIMA is highly effective for univariate time series, other methods may be preferable in different contexts. Vector Autoregression (VAR) is more suitable for multivariate forecasting, where multiple variables influence each other. Additionally, if data exhibits strong seasonality, Exponential Smoothing (ETS) or Seasonal ARIMA (SARIMA) may yield better results.

The efficient market hypothesis (Fama, 1970) suggests stock prices follow a random walk, limiting ARIMA's long-term predictive power. However, studies (Tsay, 2010) show ARIMA effectively captures short-term dependencies, particularly in less efficient markets.

In terms of exchange rate forecasting, Meese & Rogoff (1983) found ARIMA outperforms structural exchange rate models for short-term forecasting. Subsequent research (Diebold et al., 1998) confirmed its effectiveness in low-volatility environments, though accuracy declines with increased market fluctuations.

Studies by Stock & Watson (2003) and Nelson & Plosser (1982) demonstrate that ARIMA provides reliable short-term forecasts for Gross Domestic Product (GDP) and inflation. However, long-term economic shifts require alternative models, such as Vector Autoregression (VAR) and Bayesian methods.

Pindyck (1999) applied ARIMA to oil price forecasting, showing accuracy in stable periods but limitations during supply shocks. Similarly, Baur & McDermott (2010) examined gold prices, concluding that ARIMA performs well in predictable market conditions.

Despite the rise of machine learning models, ARIMA remains a cornerstone of time series forecasting due to its robust statistical framework. It is particularly valuable for short-term predictions in financial and economic applications. However, its limitations in handling nonlinearity and volatility have led to increased adoption of hybrid models, integrating ARIMA with deep learning techniques for enhanced accuracy.

2.3. Cross-validation in ARIMA Modeling

Evaluating the predictive performance of ARIMA (AutoRegressive Integrated Moving Average) models through cross-validation requires special consideration due to the sequential nature of time series data. Traditional k-fold cross-validation, which randomly partitions the dataset, is not appropriate in this context because it would violate the temporal ordering and introduce look-ahead bias. Instead, time series-specific cross-validation techniques are employed to preserve the chronological structure of the data.

In this study, cross-validation for ARIMA modeling is implemented using an expanding window approach. This procedure involves training the ARIMA model on an initial subset of the data and then incrementally expanding the training set at each step, re-estimating the model parameters, and generating out-of-sample forecasts. The process continues until the end of the dataset is reached.

More formally, for a time series $\{y_1, y_2, \dots, y_T\}$, forecasts are generated for each time point $t + h$ (where h is the forecast horizon) using a model trained on $\{y_1, \dots, y_t\}$. At each iteration, the ARIMA model is re-fitted to the growing dataset, ensuring that forecasts are always based on past data only. This approach mimics a real-time forecasting environment and provides a robust assessment of model performance under conditions of genuine data uncertainty.

Cross-validation in ARIMA modeling serves multiple purposes: it allows for the comparison of different ARIMA specifications (i.e., different combinations of autoregressive, differencing, and moving average terms), helps guard against overfitting, and produces more reliable estimates of out-of-sample accuracy. Model selection is guided by minimizing forecast error metrics such as the Mean Squared Error (MSE) or Mean Absolute Error (MAE) computed across all cross-validation steps.

2.4. Machine Learning Model

2.4.1. RNN - Recurrent Neural Networks

Recurrent Neural Networks (RNNs) are a class of artificial neural networks specifically designed to process sequential data. Unlike traditional feedforward neural networks, which treat each input independently, RNNs retain information from previous time steps, allowing them to model dependencies within time series data. This characteristic makes RNNs particularly useful for

forecasting tasks, such as predicting exchange rate movements, where past values influence future trends (Goodfellow et al., 2016).

In an RNN, each neuron receives input not only from the current time step but also from the previous hidden state, forming a recurrent connection. Mathematically, the hidden state at time t is computed as follows:

$$h_t = f(W_h h_{t-1} + W_x X_t + b) \quad (7)$$

where:

h_t represents the hidden state at time t

X_t is the input,

W_h and W_x are weight matrices,

b is the bias term,

f is non-linear activation function, such as tanh or ReLU.

The network's output is given by:

$$Y_t = W_y h_t + b_y \quad (8)$$

Where Y_t is the predicted exchange rate at time t . The model is trained using backpropagation through time (BPTT) to adjust the weight and minimize the forecasting error.

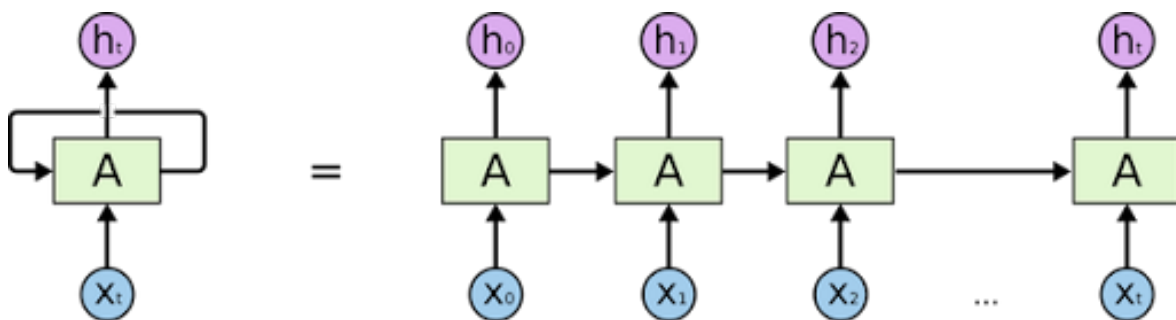


Fig. 1. Recurrent Neural Network Cell Architecture ((Ataurima Arellano, M. and Rodríguez, G., 2020))

One of the key advantages of RNNs is their ability to model time dependencies in financial data. Exchange rates are influenced by past price movements, making it essential for forecasting models to retain historical information. RNNs achieve this by propagating hidden states through time, capturing both short-term and long-term dependencies.

However, standard RNNs face challenges when modeling long sequences due to the vanishing and exploding gradient problems. As the network propagates gradients backward during training, values can either diminish to near-zero (vanishing gradient) or grow uncontrollably (exploding gradient), leading to unstable learning (Hochreiter & Schmidhuber, 1997). These issues limit the ability of RNNs to capture long-range dependencies in exchange rate data.

To mitigate these challenges, more advanced architectures, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), introduce gating mechanisms that selectively retain or discard information. However, even in its basic form, an RNN remains a fundamental approach to sequential modeling and serves as the foundation for more sophisticated deep learning techniques.

The RNN model is trained using stochastic gradient descent (SGD) with adaptive optimizers such as Adam, which dynamically adjusts learning rates to improve convergence. The loss function is typically Mean Squared Error (MSE), measuring the deviation between actual and predicted exchange rates.

The network is trained over multiple epochs, with batch processing used to improve computational efficiency. Hyperparameters such as the number of hidden units, learning rate, and sequence length are fine-tuned through experimentation.

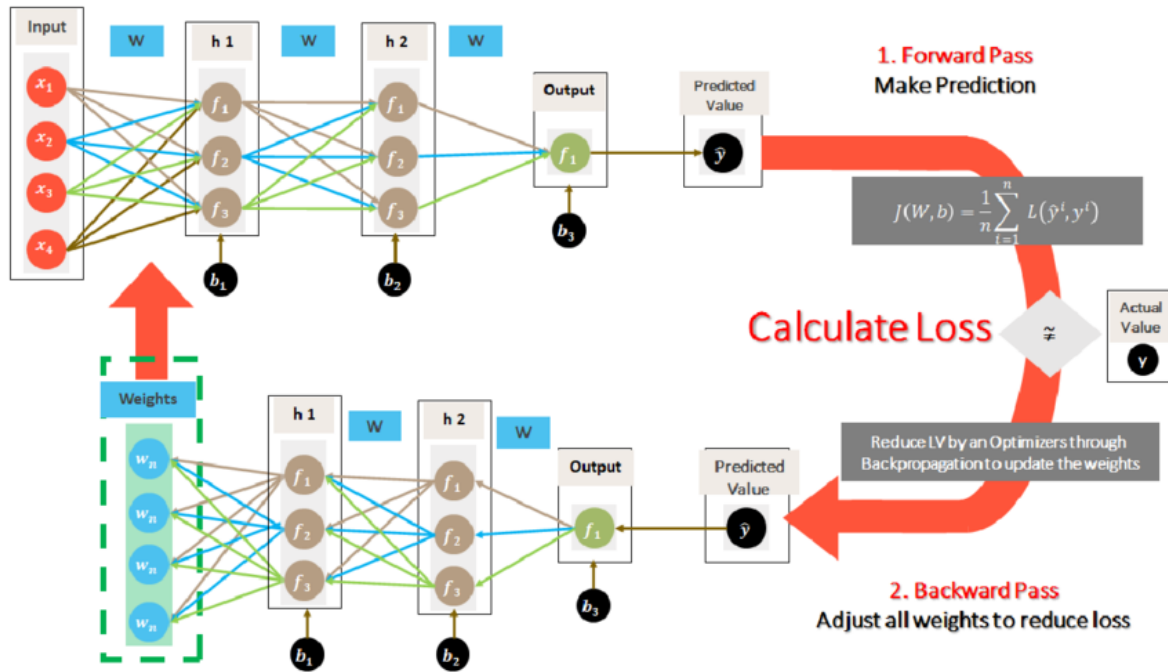


Fig. 2. Recurrent neural network overall training process (110. Al-Selwi S., Hassan M., Abdulkadir S., Muneer A. (2023))

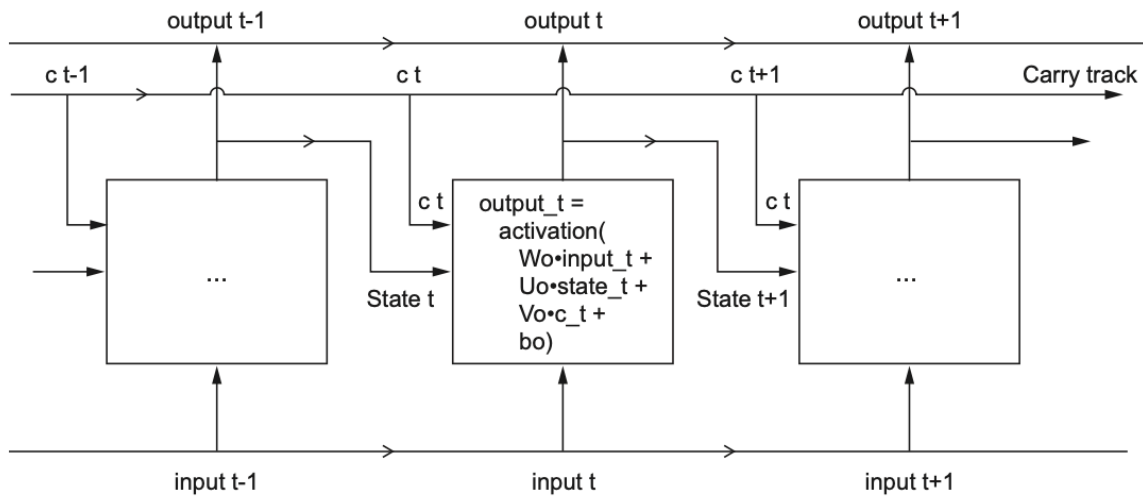
2.4.2. LSTM - Long Short-Term Memory networks

Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN), were first introduced by Hochreiter and Schmidhuber in 1997 to address the vanishing gradient problem inherent in traditional RNNs. LSTMs excel at capturing long-range dependencies in sequential data, which makes them particularly effective for time series forecasting (Hochreiter & Schmidhuber, 1997). In François Chollet's Deep Learning with Python (2018), the author emphasizes how LSTMs are adept at learning from sequences where the timing and context of data points are crucial, such as in speech recognition, natural language processing, and time series forecasting.

LSTMs work by maintaining a memory cell, which helps the model retain information for long periods, and by using gates (input, forget, and output) to control the flow of information. This architecture makes them highly suitable for tasks involving sequential data, such as financial time series or exchange rate forecasting, where past values can have significant predictive power for future trends.

Figure 3 illustrates the architecture of an LSTM model used for sequence prediction. At each time step, the carry values, denoted as C_t , play a crucial role

in the LSTM cell's operations. These carry values interact with both the input and recurrent connections, undergoing a dense transformation. This transformation involves a dot product with a weight matrix, followed by adding a bias term and applying an activation function. The carry information also influences the state passed to the next time step, where it is adjusted through a multiplication operation and an activation function. In summary, the carry data flow serves to modulate both the output and the state that are carried forward to the subsequent time step (as shown in Figure 3).



Fir. 3. From a SimpleRNN to an LSTM: adding a carry track (Chollet, F. (2018))

LSTMs have been widely used in forecasting financial time series, including exchange rates. Several studies have demonstrated their superior performance over traditional time series models like ARIMA and VAR. For instance, a study by Fischer & Krauss (2018) applied LSTM networks to predict stock prices and found that they outperformed classical machine learning methods. Similarly, Zhang et al. (2020) utilized LSTM to forecast exchange rates, achieving higher accuracy than models such as ARIMA and GRU (Gated Recurrent Unit).

2.5. Evaluation Metrics

2.5.1. MSE - Mean Squared Error

The Mean Squared Error (MSE) is a widely used performance metric in forecasting models, particularly for assessing the accuracy of continuous predictions. MSE measures the average squared difference between actual

values and predicted values, penalizing larger errors more significantly than smaller ones. The formula for MSE is given by: The Mean Squared Error (MSE) is a widely used performance metric in forecasting models, particularly for assessing the accuracy of continuous predictions. MSE measures the average squared difference between actual values and predicted values, penalizing larger errors more significantly than smaller ones. The formula for MSE is given by:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (9)$$

where:

y_i represents the actual observed value at time i ,

$\hat{y}_i$ is the corresponding predicted values,

n is the total number of observations.

Since MSE squares the error terms, it emphasizes larger deviations, making it particularly useful when substantial forecasting errors need to be minimized. However, its sensitivity to outliers means that extreme values can disproportionately influence the overall metric.

2.5.2. Diebold-Mariano (DM) Test

This section outlines the methodological approach for evaluating forecast accuracy using the Diebold-Mariano (DM) test. The test provides a formal statistical framework to assess whether the predictive performance of two competing forecasting models differs significantly.

Forecast evaluation plays a central role in empirical modeling, particularly when multiple methods are used to predict the same variable. Simple comparisons of mean squared error (MSE) or mean absolute percentage error (MAPE) offer descriptive insights but do not indicate whether observed differences in performance are statistically meaningful. The Diebold-Mariano test, introduced by Diebold and Mariano (1995), addresses this gap by testing the null hypothesis of equal predictive accuracy.

Let $\{e_{1,t}\}$ and $\{e_{2,t}\}$ represent forecast errors from two models over a forecast evaluation period of length T . The test begins by constructing a loss differential series:

$$d_t = L(e_{1,t}) - L(e_{2,t}) \quad (10)$$

where $L(\cdot)$ is a chosen loss function, commonly the squared error $L(e_t) = e_t^2$, though absolute error or other forms may also be used depending on the focus of the analysis.

The null hypothesis of equal forecast accuracy is expressed as:

$$H_0: \mathbb{E}[d_t] = 0 \quad (11)$$

This is tested against the alternative that the expected loss differential is nonzero, implying that one model systematically outperforms the other.

The Diebold-Mariano test statistic is defined as:

$$DM = \frac{\bar{d}}{\sqrt{\frac{\hat{\gamma}(0)}{T} + 2 \sum_{k=1}^{h-1} \hat{\gamma}(k)/T}} \quad (12)$$

where $\bar{d}$ is the sample mean of the loss differential, $\hat{\gamma}(k)$ denotes the sample autocovariance of d_t at lag k , and h is the forecast horizon. The denominator accounts for potential autocorrelation in the loss differential series, particularly relevant for multi-step-ahead forecasts.

Under the null hypothesis and regularity conditions, the DM statistic converges in distribution to the standard normal, allowing inference through conventional critical values. For finite samples, a small-sample correction proposed by Harvey, Leybourne, and Newbold (1997) is often applied to improve test size.

Several factors influence the outcome and interpretation of the DM test:

1. Choice of loss function: The test results can vary depending on whether squared error, absolute error, or another criterion is used.
2. Forecast horizon: The test is sensitive to the length of the forecast window, particularly in the presence of serial correlation.

3. Sample size: With short evaluation periods, the DM test may lack power, underscoring the importance of using corrected versions for small samples.

Despite these limitations, the DM test remains a standard benchmark for evaluating forecast performance in economics and finance. It offers a transparent and statistically grounded mechanism to determine whether differences in model accuracy are due to chance or reflect genuine predictive superiority.

2.5.3. Giacomini-White (GW) test

This section presents the framework for comparing the predictive performance of conditional forecasting models using the Giacomini-White (GW) test. This test extends the Diebold-Mariano framework by evaluating forecast accuracy in a context where information sets differ or time-varying features are explicitly modeled.

While traditional forecast comparison tests, such as the Diebold-Mariano test, are valuable for assessing overall accuracy, they assume that the forecast evaluation environment is static and that forecasts are generated under identical conditions. In many applied settings, particularly in macroeconomic and financial forecasting, models are estimated recursively, and the underlying data-generating process may evolve over time. The Giacomini-White test (Giacomini & White, 2006) addresses this by comparing conditional predictive performance, taking into account model dynamics and structural features.

Let $e_{1,t+h}$ and $e_{2,t+h}$ denote the h -step-ahead forecast errors from two competing models, where t indexes the time at which the forecast is made. A general loss function $L(\cdot)$ is applied to these errors to construct the loss differential:

$$d_t = L(e_{1,t+h}) - L(e_{2,t+h}) \quad (13)$$

Unlike unconditional forecast evaluation, the GW test examines whether the expectation of this loss differential, conditional on information available at forecast origin, is equal to zero:

$$H_0: \mathbb{E}[d_t | \mathcal{F}_t] = 0 \quad (14)$$

Here, $\mathcal{F}_t$ represents the information set at time t . Under the null hypothesis, both models are equally accurate given the available information at the time of forecasting.

The Giacomini-White test proceeds by regressing the loss differential d_t on a set of instruments, typically chosen to reflect features known at the forecast origin (such as lagged variables or volatility proxies):

$$d_t = z_t^\top \beta + u_t \quad (15)$$

The null hypothesis $H_0: \beta = 0$ implies that the loss differential cannot be systematically explained by available information, and hence neither model consistently outperforms the other in a conditional sense.

The test statistic is constructed as a Wald-type test:

$$GW = T \cdot \hat{\beta}^\top \left(\widehat{Var}(\hat{\beta}) \right)^{-1} \hat{\beta} \quad (16)$$

where T is the number of forecasts and $\widehat{Var}(\hat{\beta})$ is a robust estimate of the covariance matrix of the estimated coefficients. Under the null hypothesis, this statistic follows a chi-squared distribution with degrees of freedom equal to the number of instruments.

The GW test offers several advantages over standard accuracy tests:

- Time-varying evaluation: It is sensitive to structural changes and regime shifts, which are common in real-world forecasting contexts.
- Conditional framework: It accounts for the forecast environment, allowing for richer diagnostic analysis.
- Flexibility: Users can customize the instrument set to reflect specific economic theories or empirical features.

In practice, the GW test is particularly useful when comparing models that differ in their responsiveness to new information or structural dynamics—such as comparing traditional time series models with adaptive machine learning algorithms.

3. Empirical Results

3.1. Data Explanation and Preparation

This study utilizes Python to analyze two datasets containing daily exchange rate information for five currency pairs: EUR/USD, EUR/GBP, EUR/JPY, USD/RUB, and AUD/NZD, sourced from Yahoo Finance. The daily dataset, which spans from January 3, 2000, onward, provides long-term historical data. These currency pairs were chosen due to their high liquidity and economic significance in global financial markets.

The daily exchange rate dataset contains 11,976 daily observations. The timestamp column (ts) contains 11,976 non-null entries and serves as the index for temporal alignment. However, not all currency series are complete:

EURJPY=X has 1,446 missing entries
EURGBP=X has only 19 missing entries
EURUSD=X has 1,889 missing entries
USDRUB=X has the largest gap, with 2,102 missing values
AUDNZD=X is missing 1,851 observations

These discrepancies likely arise from differences in market operating days and data availability across time zones and regions. Handling these missing values is essential before any modeling can be performed. In this study, linear interpolation (1) is applied to fill gaps, maintaining temporal consistency while avoiding the introduction of artificial volatility.

The descriptive statistics of the numerical columns reveal several insights into the behavior of the exchange rates:

The EUR/JPY series has a wide range (94.21 to 169.55), with a mean of approximately 131.34 and a relatively high standard deviation (14.51), indicating noticeable volatility over the sample period.

EUR/GBP is more stable, with a mean of 0.7855 and a narrower standard deviation of 0.0969, consistent with the behavior of major currencies in developed markets.

EUR/USD also shows a moderate standard deviation (0.1263) around a mean of 1.2390, which aligns with its historically observed range.

The USD/RUB series exhibits significant volatility, with values ranging from 0.72 to nearly 139.00 and a high standard deviation of 20.13. This reflects macroeconomic instability and geopolitical risk associated with the Russian ruble.

AUD/NZD appears relatively stable, ranging between 1.00 and 1.37, with a mean close to 1.14 and low standard deviation (0.0853), consistent with the close economic ties and similar monetary policy frameworks between Australia and New Zealand.

Dataset summary

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11976 entries, 0 to 11975
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0    ts          11976 non-null  object
1    EURJPY=X    10530 non-null  float64
2    EURGBP=X    11957 non-null  float64
3    EURUSD=X    10087 non-null  float64
4    USDRUB=X    9874 non-null   float64
5    AUDNZD=X    10125 non-null  float64
dtypes: float64(5), object(1)
memory usage: 561.5+ KB
None
```

Missing values in each column:

```
ts          0
EURJPY=X    1446
EURGBP=X     19
EURUSD=X    1889
USDRUB=X    2102
AUDNZD=X    1851
dtype: int64
```

Descriptive statistics of numerical columns:

	EURJPY=X	EURGBP=X	EURUSD=X	USDRUB=X
AUDNZD=X				
count	10530.000000	11957.000000	10087.000000	9874.000000
mean	131.340757	0.785502	1.238998	46.916704
std	14.512268	0.096869	0.126269	20.130811
	0.085263			

min	94.209999	0.573000	0.959619	0.716200	
1.002900					
25%	122.664001	0.688000	1.129331	29.059999	
1.071500					
50%	131.246994	0.825650	1.227144	34.502850	
1.100688					
75%	138.537495	0.867090	1.331851	64.680000	
1.205100					
max	169.550003	0.979000	1.598798	138.965103	1.374900

In financial time series data, extreme values — often referred to as spikes or outliers — can distort model estimates and reduce forecasting accuracy. This issue is particularly relevant for emerging market currencies like the Russian ruble (USD/RUB), which may experience abrupt shifts due to geopolitical or macroeconomic shocks. To mitigate this, an outlier detection and correction procedure was applied to the USD/RUB series.

The method uses a rolling Z-score approach, where each point in the series is compared to a centered rolling window of size 7.

This approach maintains the general structure of the time series while effectively reducing the influence of extreme events that may not reflect underlying trends. The cleaned USD/RUB series is thus better suited for time series modeling and contributes to improved forecasting performance.

3.2. Data Visualization

Figure 4 presents the historical daily price levels of six exchange rates — EURJPY, EURGBP, EURUSD, USDRUB, and AUDNZD — spanning from early 2019 to 2023. These time series exhibit distinct dynamics reflecting global macroeconomic events, monetary policy changes, geopolitical developments, and market sentiment.

EURJPY=X: The EUR/JPY pair shows a steady upward trend beginning in mid-2020, suggesting persistent euro strength or yen weakness. After a relatively flat period from 2019 to 2020, the exchange rate climbs significantly, peaking in late 2023 near 160.

EURGBP=X: The EUR/GBP pair exhibits mean-reverting behavior within a bounded range. Notable volatility is observed in 2020 and 2022, including a

sharp spike — possibly reflecting Brexit-related developments or market uncertainty during the Russia–Ukraine conflict.

EURUSD=X: This pair trends downward from 2019 to mid-2022, reflecting USD appreciation or euro depreciation. A temporary recovery occurs in 2020–2021, peaking above 1.20, followed by a continued decline and moderate stabilization post-2022.

USDRUB=X: The USD/RUB rate is notably volatile, especially from early 2022 onward, where extreme price fluctuations align with geopolitical tensions—most likely due to sanctions and financial market disruptions following Russia’s invasion of Ukraine. The spike above 130 and subsequent collapse indicate high instability.

AUDNZD=X: This cross remains relatively stable compared to others, fluctuating within a narrow range between 1.03 and 1.13. Minor cyclical patterns suggest limited directional trend, aligning with the economic correlation between Australia and New Zealand.

This visualization highlights the heterogeneity in exchange rate behavior across pairs, underlining the importance of model selection and forecasting methodology tailored to each currency's characteristics.

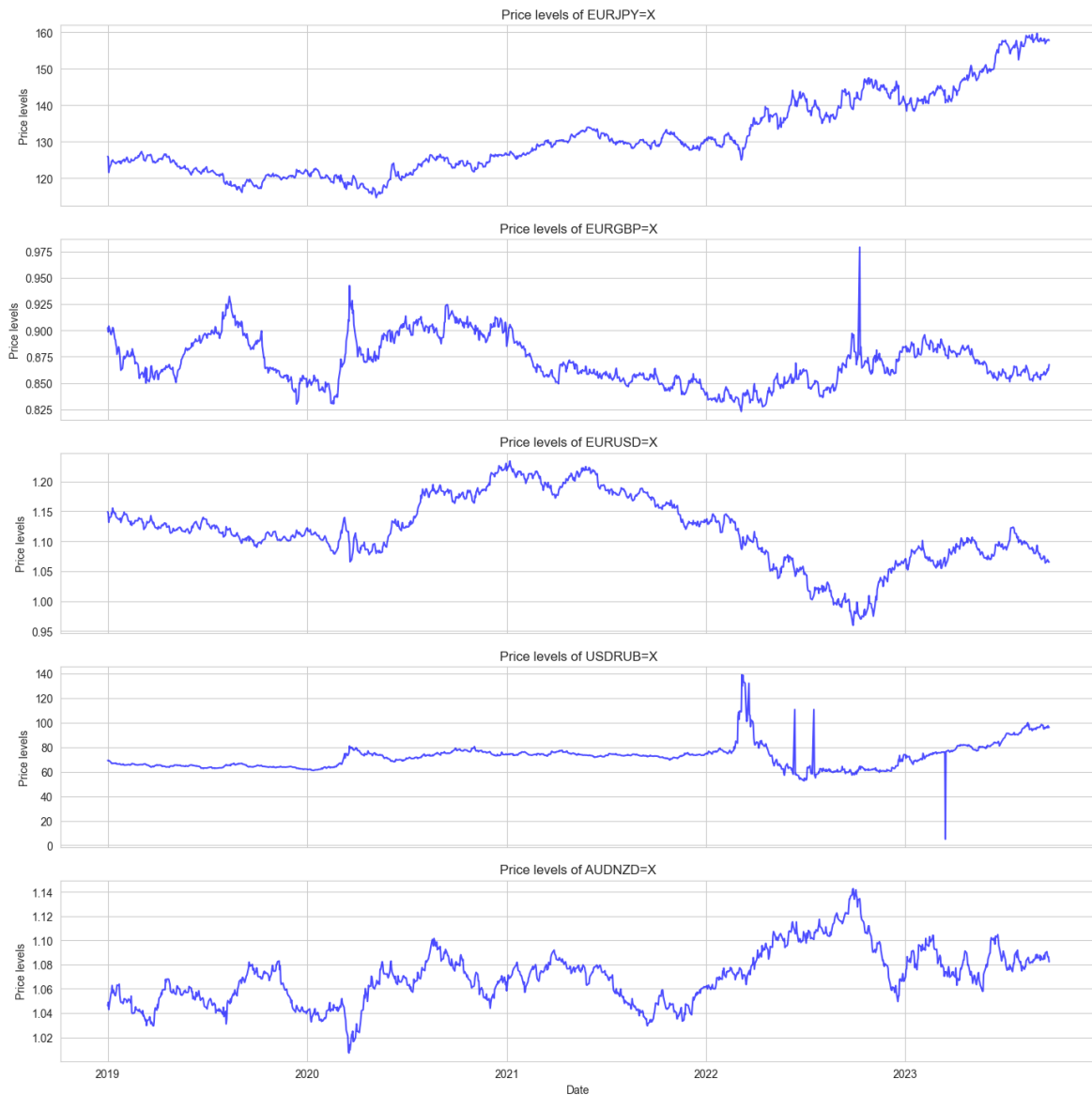


Fig. 4. Price levels of exchange rates from 2019 – 2023 years

Figure 5 displays the daily log returns of five major exchange rates from 2019 to 2023: EURJPY, EURGBP, EURUSD, USDRUB, and AUDNZD. Log returns provide a stationary time series representation of price movements and are more suitable for forecasting and volatility modeling due to their additive properties and relative scale invariance.

EURJPY=X: The log returns of EUR/JPY remain relatively stable until 2020, after which increased volatility is observed, particularly through 2022–2023. This suggests heightened uncertainty or fundamental shifts in the euro or yen economies during this period.

EURGBP=X: This series exhibits spikes in volatility in mid-2020 and again in early 2022, likely driven by Brexit aftershocks and geopolitical disruptions. Overall, the returns remain within a narrow band, reflecting a tightly managed or co-integrated relationship between the euro and the pound.

EURUSD=X: The EUR/USD log return series reflects multiple regime shifts. It shows moderate volatility with peaks during the COVID-19 pandemic and rising USD strength through 2022. Some stabilization is observed in late 2023.

USDRUB=X: This series is marked by extreme volatility, particularly around early 2022, corresponding to the Russia–Ukraine conflict and subsequent international sanctions. Spikes and drops of abnormal magnitude indicate a highly unstable market environment for the Russian ruble.

AUDNZD=X: The log returns of AUD/NZD display low volatility and a range-bound pattern, consistent with the economic and geographical proximity of Australia and New Zealand. No large shocks are apparent, making this pair relatively more stable for modeling.

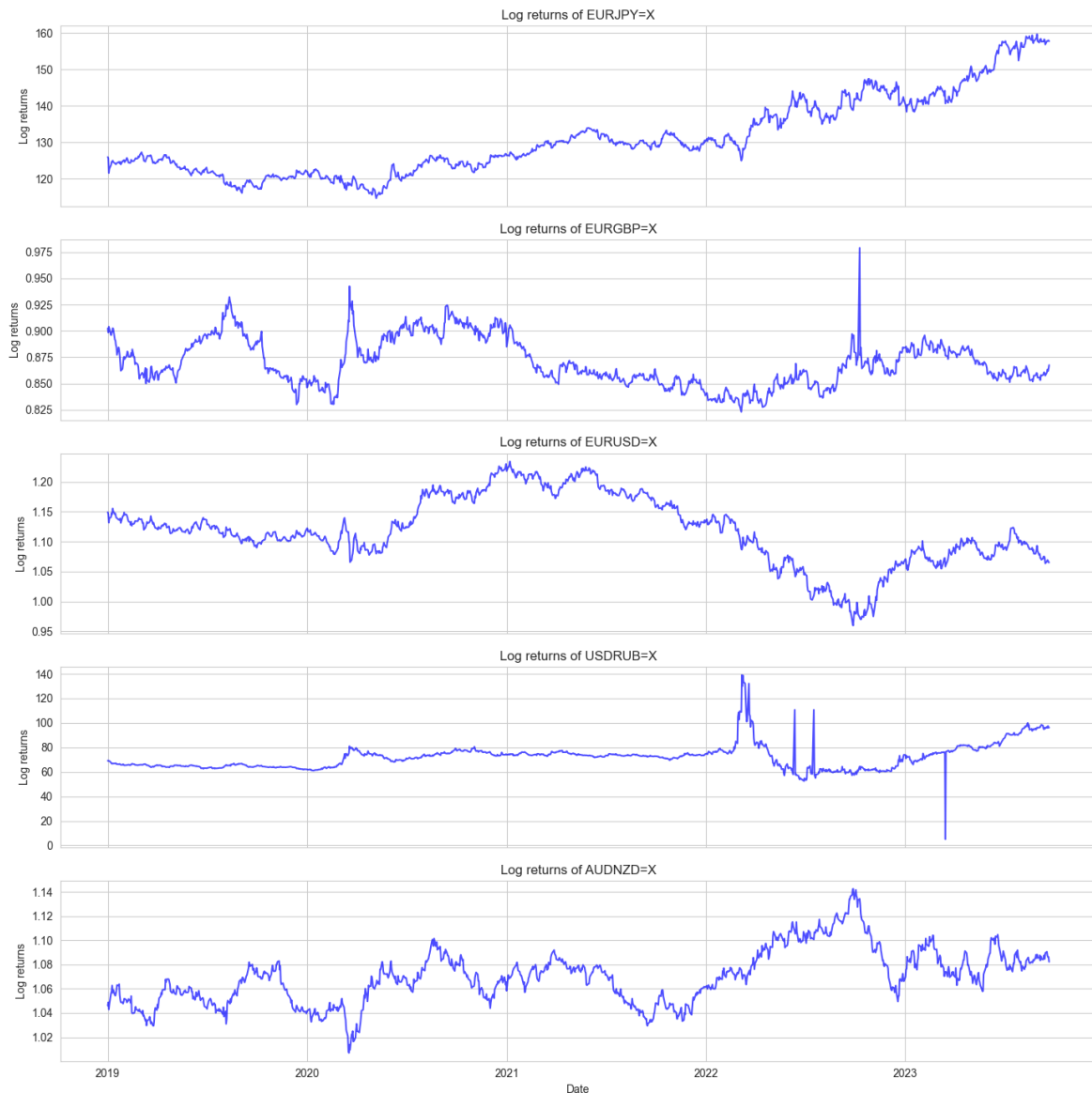


Fig. 5. Log returns of exchange rates from 2019 – 2023 years

Figure 6 shows five time series plots, each representing squared log returns for different currency pairs over time.

EURJPY=X (Euro/Japanese Yen)

The chart shows periods of low volatility punctuated by occasional spikes, indicating sudden increases in return volatility. Notable spikes occur around 2020 and again in 2022–2023.

EURGBP=X (Euro/British Pound)

Generally low volatility, with a significant spike around late 2022, which may correspond to a specific economic or political event impacting the Euro or Pound.

3. EURUSD=X (Euro/US Dollar)

More frequent small spikes in volatility compared to the others. A noticeable increase in volatility around 2021–2022.

4. USDRUB=X (US Dollar/Russian Ruble)

Very low and stable for most of the time period. A massive spike occurs around early 2022, likely due to geopolitical events (e.g., Russia-Ukraine conflict), reflecting extreme volatility.

5. AUDNZD=X (Australian Dollar/New Zealand Dollar)

Exhibits regular, moderate volatility throughout the time period. The pattern is more consistent with repeated small to moderate spikes.

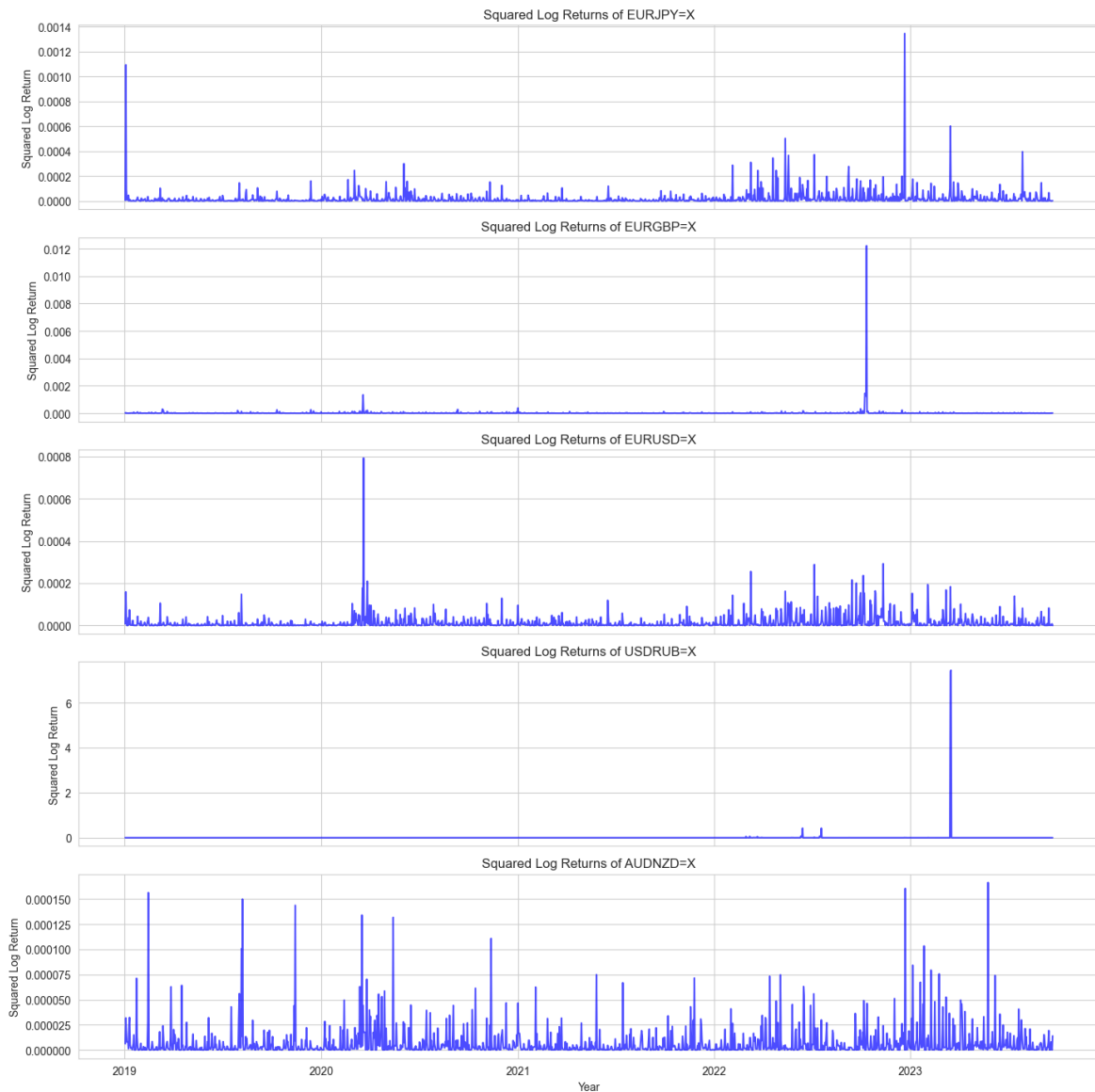


Fig. 6. Squared log returns of exchange rates from 2019 – 2023 years

Figure 7 complements the first one by showing the price levels (not volatility) for the same currency pairs, over the same time period (~2019 to 2023), with 30-day rolling averages overlaid.

EURJPY=X (Euro/Japanese Yen)

General upward trend from mid-2020 through 2023. The 30-day moving average (black dashed line) smooths out short-term noise but follows the overall upward movement.

EURGBP=X (Euro/British Pound)

Volatile and range-bound, moving roughly between 0.83 and 0.93. Notable spikes align with the same volatility seen in the previous squared returns chart, especially around late 2022.

EURUSD=X (Euro/US Dollar)

Downward trend from 2019 to 2022, followed by a partial recovery.

The 30-day average tracks closely, showing a smooth version of this trend.

USDRUB=X (US Dollar/Russian Ruble)

Mostly flat until early 2022, then an explosive surge during the Russia-Ukraine conflict. Very large spikes and anomalies (e.g., sudden drops) reflect instability during sanctions or market suspensions.

AUDNZD=X (Australian Dollar/New Zealand Dollar)

Shows cyclical behavior with moderate fluctuation in a narrow range (~1.02 to 1.14). Rolling average effectively captures the cycles and shows relatively less directional movement.

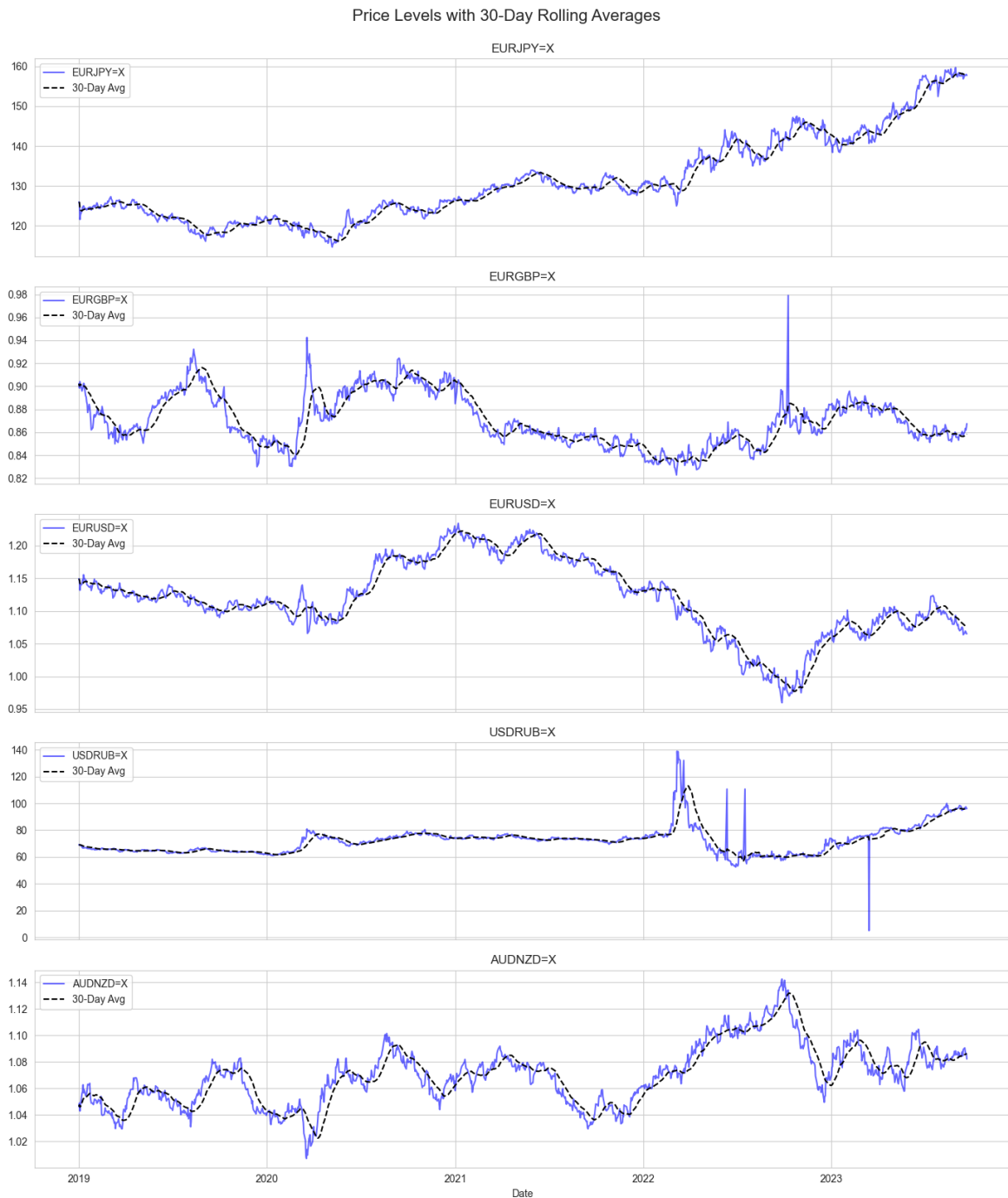


Fig. 7. Price levels with 30-day rolling averages

Figure 8 presents log returns with 30-day rolling averages for the same five currency pairs as before, across the same timeframe (2019–2023).

EURJPY=X

Most returns are between -0.02 and 0.02, with occasional larger negative spikes. Rolling average remains close to zero, indicating no strong

upward/downward return trend, though volatility seems to increase from 2022 onward.

EURGBP=X

Very stable log returns, close to zero for most of the period. One sharp negative return around 2022 aligns with a spike seen in the squared returns and price charts — likely a market shock.

EURUSD=X

Slightly wider spread of returns than EURGBP. Occasional sharp drops (e.g., early 2020, late 2021) reflect external shocks or news events.

USDRUB=X

Almost flat until early 2022. Then, extreme outliers in both positive and negative directions — likely due to geopolitical turmoil (e.g., Russia-Ukraine war), sanctions, or market disruptions.

The rolling average briefly diverges from zero during that time.

5. AUDNZD=X

Log returns are relatively consistent and centered around zero. Slightly more volatility around 2020 and 2022, which aligns with general global market uncertainty during COVID and post-pandemic adjustments.

Most currency pairs have log returns centered around zero, suggesting no strong long-term drift.

Periods of extreme returns align with the spikes in squared returns and large price moves in the previous charts.

USDRUB=X is the most volatile, especially post-2022.

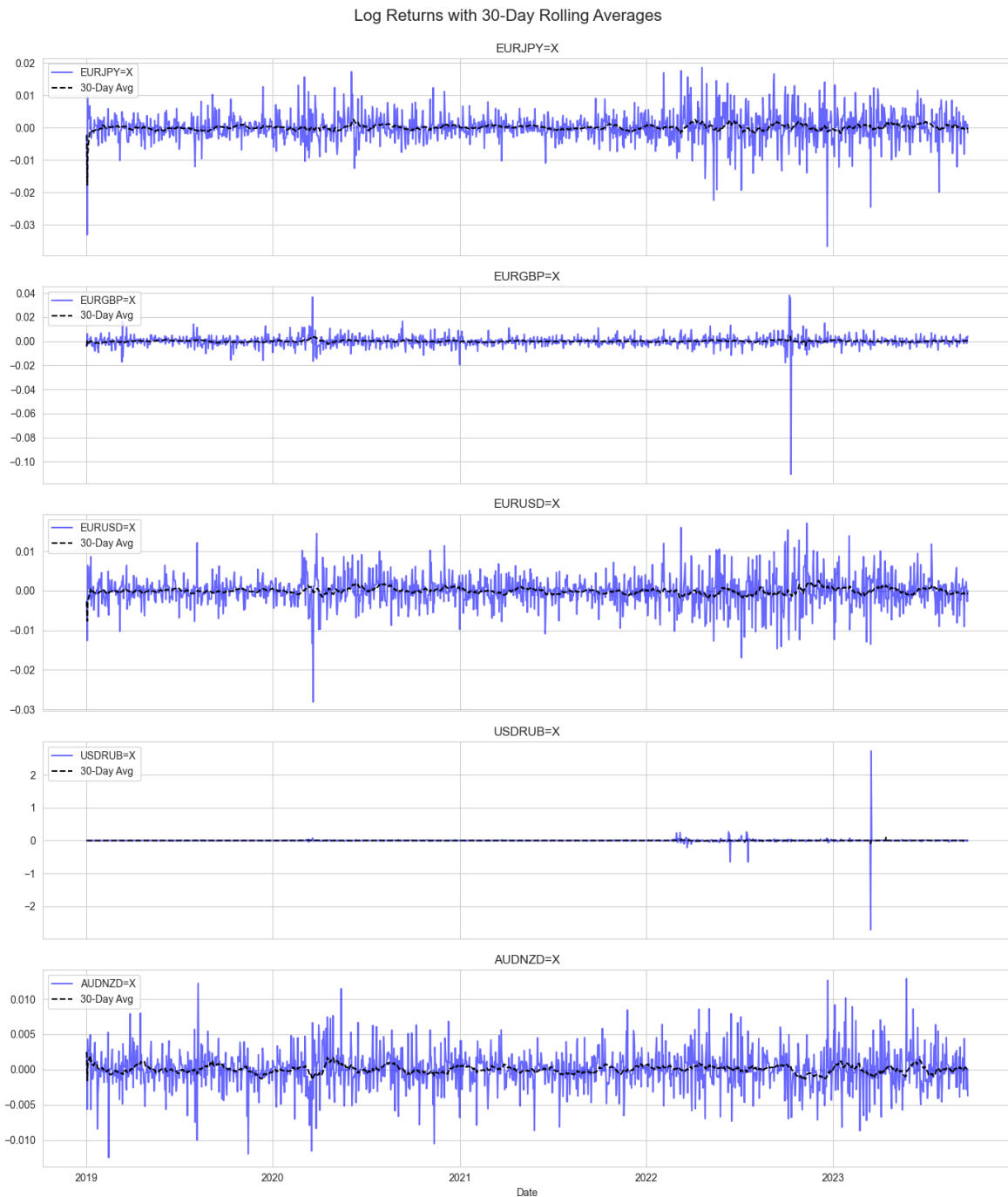


Fig. 8. Log returns with 30-day rolling averages

Figure 9 visualizes squared log returns of five currency pairs over time, along with their 30-day rolling averages.

EURJPY=X:

Spikes in volatility early on (2018–2019) and periodically after that. Overall relatively low and stable squared returns after 2020.

EURGBP=X:

Very low volatility, with a noticeable spike around late 2022.

EURUSD=X:

A significant spike in early 2020 (likely related to COVID-19 market turbulence). Subsequent small but frequent volatility upticks, particularly from 2021 onward.

USDRUB=X:

Extreme spike in early 2022 — a very prominent outlier (likely reflecting geopolitical tensions, i.e., the Russia-Ukraine conflict). Otherwise, relatively low and calm squared returns.

AUDNZD=X:

Consistent small fluctuations, with many short-lived spikes, suggesting more frequent but lower-amplitude volatility.

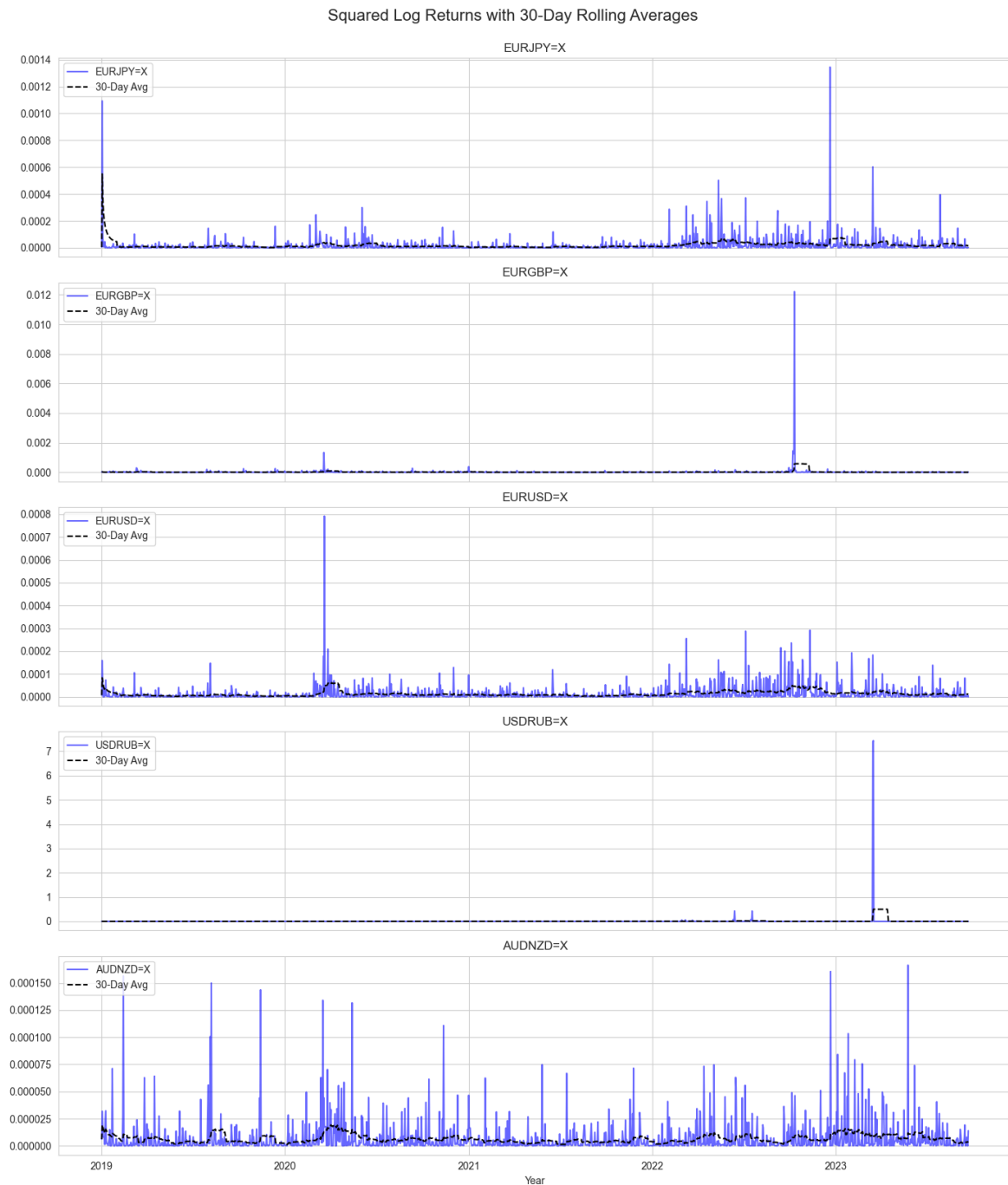


Fig. 9. Squared log returns with 30-day rolling averages

Figure 10 presents the classical time series decomposition of exchange rate price levels for five currency pairs, separating each into three core components: trend (long-term progression); seasonal (regular periodic fluctuations); residual (random noise or irregular patterns).

1. EURJPY=X

Trend: Strong and steady upward movement, especially post-2020.

Seasonal: Clear yearly seasonality, sharp cycles.

Residual: Random noise, spikes notably in 2020–2022, but relatively narrow band.

2. EURGBP=X

Trend: U-shaped pattern (decline into 2022, then recovery).

Seasonal: Strong yearly patterns, consistent amplitude.

Residual: Low-level irregularities, no major outliers.

3. EURUSD=X

Trend: Rise into 2021, followed by a marked decline into 2022, slight recovery thereafter.

Seasonal: Clear cyclic pattern, typical of exchange rates.

Residual: Higher variability from 2020–2022, some persistent deviation from the expected pattern.

4. USDRUB=X

Trend: Highly volatile, reflects geopolitical shocks (e.g., big jumps in 2022).

Seasonal: Weak and irregular—less consistent than others.

Residual: Large spikes and unstable behavior post-2022, indicating model doesn't fully capture volatility.

5. AUDNZD=X

Trend: Several waves of increase and decrease; less directional.

Seasonal: Well-defined, small-magnitude cycles.

Residual: Moderate noise with no huge outliers, fairly stable.

Classical Decomposition for Price Levels of Exchange Rates (Trend, Seasonal, Residual)

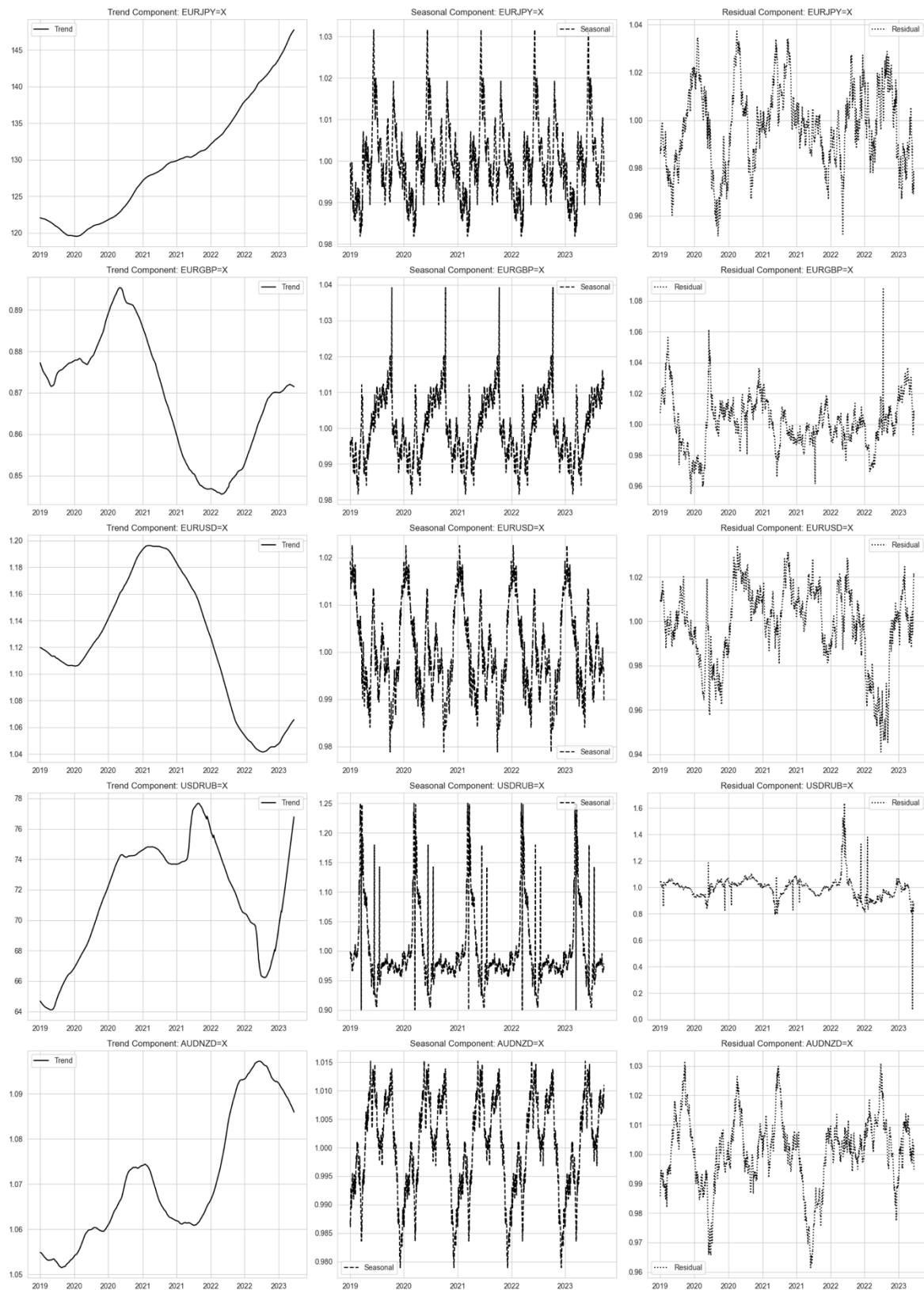


Fig. 10. Classical decomposition for price levels of exchange rates

Figure 11 is STL decomposition (Seasonal-Trend-Loess) for the five exchange rate.

Trend Components (STL)

EURJPY=X: The trend shows a consistent and strong upward movement from 2020 onward, with steady appreciation of the euro against the yen. The increase is smoother compared to the classical decomposition.

EURGBP=X: This trend follows a U-shaped path — declining into early 2022, then reversing direction and rising through 2023. The trend is more refined and less noisy than in classical decomposition.

EURUSD=X: Shows a clear rise into 2021, a steady decline into 2022, and a modest rebound in 2023. The STL method highlights this structure with less distortion than the classical approach.

USDRUB=X: The STL trend captures a clear upward trajectory with fluctuations. Notably, the sharp rise in 2022 is emphasized, reflecting geopolitical impacts and market volatility.

AUDNZD=X: The trend follows a wave-like pattern, with upward momentum at the end of the period. Changes are captured more smoothly, avoiding overfitting to short-term fluctuations.

Seasonal Components (STL)

EURJPY=X: Seasonal variation is prominent, with increasing amplitude toward the end of the period — indicating stronger cyclical forces in recent years.

EURGBP=X: Seasonal patterns are subtle but evident, with moderate increases in intensity around 2022–2023. There's a notable spike likely reflecting short-term events interpreted as seasonal.

EURUSD=X: Displays low-magnitude seasonal behavior. Seasonal fluctuations are present but weak and relatively consistent, suggesting a less dominant seasonal component.

USDRUB=X: The seasonal component is highly volatile and irregular, unlike other pairs. It contains large spikes, especially during and after 2022, indicating that STL attributes some of the price shocks to seasonality.

AUDNZD=X: Displays a consistent and clear seasonal pattern, with small but recurring oscillations — similar in structure to what was seen in classical decomposition, but slightly more detailed.

Residual Components (STL)

EURJPY=X: Residuals remain fairly centered around zero but show bursts of volatility, particularly in 2020–2022. These represent noise or events not captured by trend or seasonality.

EURGBP=X: Residuals are tightly clustered around zero, indicating a strong model fit. Small spikes do appear during more volatile periods but are not extreme.

EURUSD=X: The residual series is fairly stable, with deviations becoming more pronounced from mid-2022 onward. These are likely related to macroeconomic shocks.

USDRUB=X: Residuals are very large and erratic — clearly capturing the shocks that STL couldn't attribute to either trend or seasonality. Especially after 2022, these residuals dominate the series.

AUDNZD=X: Residuals are moderate, mostly centered around zero with small, scattered deviations. Overall, this reflects a good fit between observed data and the STL model.

STL Decomposition for Price Levels of Exchange Rates (Trend, Seasonal, Residual)

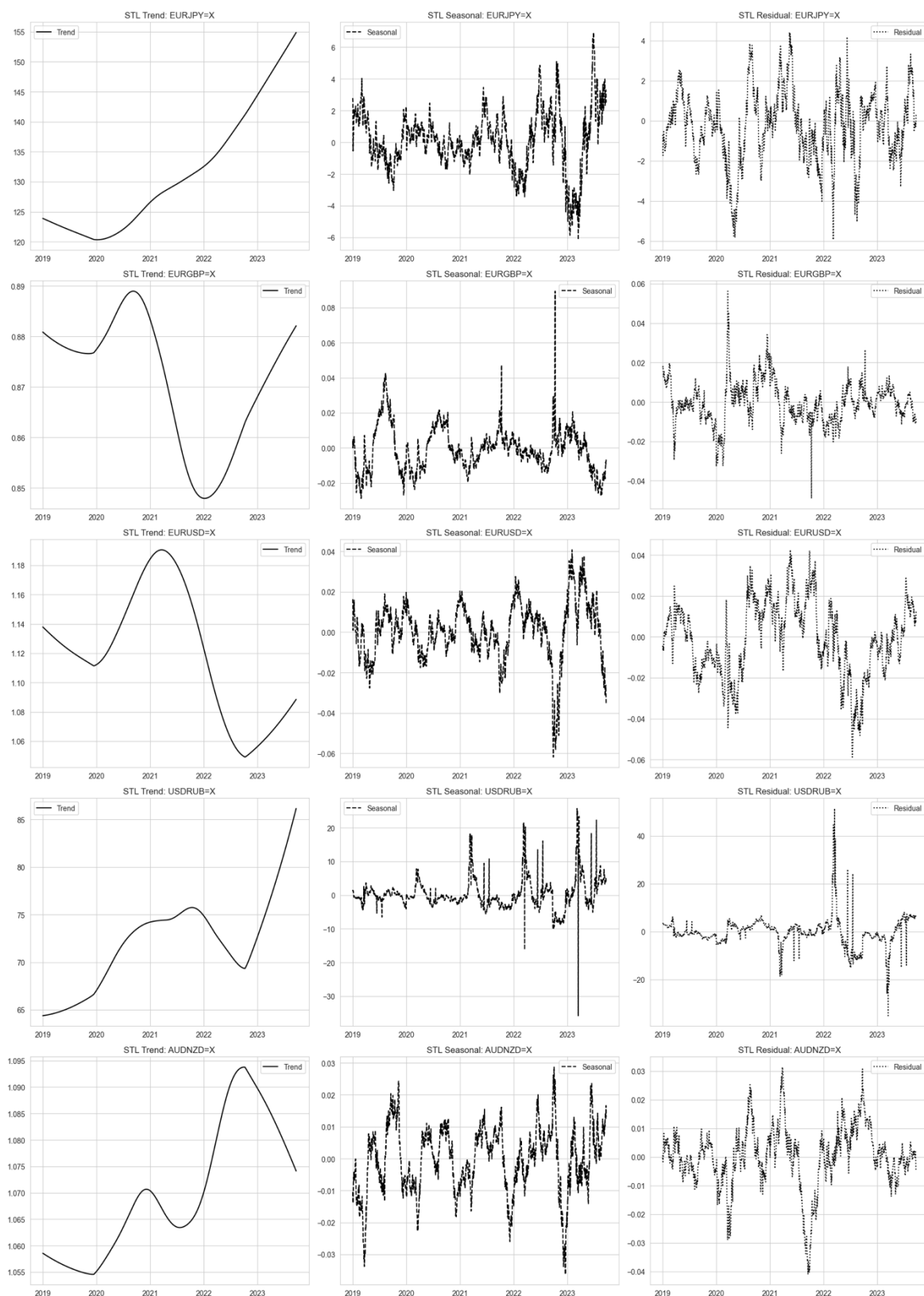


Fig. 11. STL decomposition for price levels of exchange rates

Figure 12 shows classical time series decomposition applied to the log returns of exchange rates for various currency pairs.

1. EUR/JPY

Trend: Gradually rising with cyclical movements.

Seasonal: Minor seasonal effects with regular but small oscillations.

Residual: Fluctuates around zero, showing moderate noise.

2. EUR/GBP

Trend: Volatile with peaks in 2020 and 2022, declining after that.

Seasonal: Stronger than EUR/JPY; clear recurring patterns.

Residual: Contains significant spikes (e.g., volatility events).

3. EUR/USD

Trend: Clear U-shape — strong decline during COVID-19, with recovery afterward.

Seasonal: Noticeable small seasonal oscillations.

Residual: Mostly stable with brief extreme deviations (e.g., March 2020 pandemic shocks).

4. USD/RUB

Trend: Drastic shifts (e.g., in 2022 due to geopolitical shocks), large volatility.

Seasonal: Outliers dominate; not a stable seasonal pattern.

Residual: Very high spikes, showing extreme events (likely from sanctions or war impacts).

5. AUD/NZD

Trend: Similar to EUR/JPY — cyclical, peaking in 2021 and declining after.

Seasonal: Clear, narrow seasonal band.

Residual: Moderate noise with occasional outliers.

Classical Decomposition for Log Returns of Exchange Rates (Trend, Seasonal, Residual)

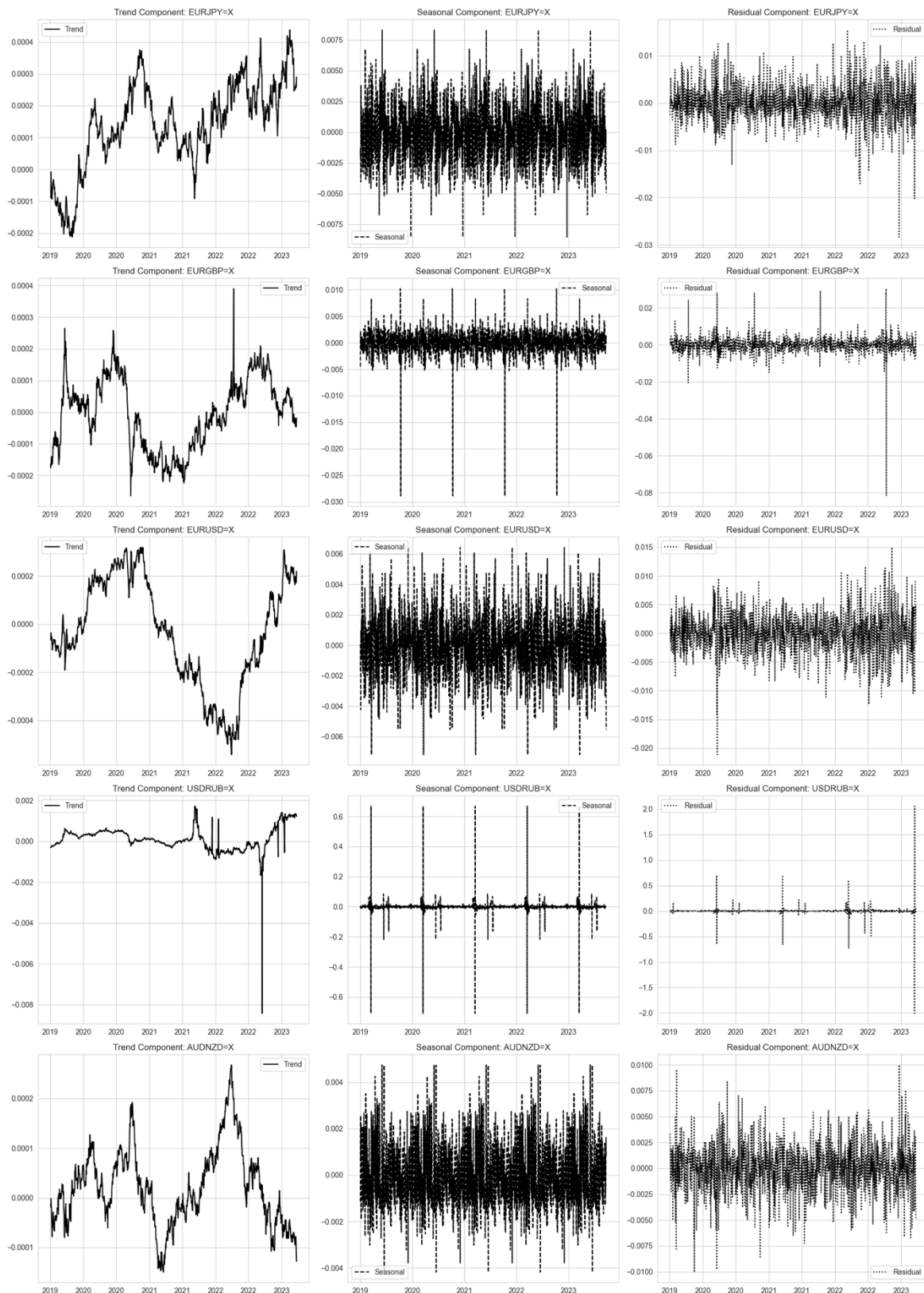


Fig. 12. Classical decomposition for log returns of exchange rates

Figure 13 presents an STL decomposition (Seasonal-Trend decomposition using Loess) for the log returns of exchange rates, breaking each series into Trend, Seasonal, and Residual components.

1. EUR/JPY

Trend: Smooth variations with noticeable cycles and peaks — STL captures short-term shifts.

Seasonal: Slight cyclical fluctuations, narrower than classical decomposition.

Residual: Less dispersed; noise is well-isolated and lower in magnitude.

2. EUR/GBP

Trend: Shows smoother movements compared to classical version—more local trends captured.

Seasonal: Sharp changes in late 2022–2023 (possibly due to Brexit-related or economic shocks).

Residual: Relatively small fluctuations, outliers clearly separated.

3. EUR/USD

Trend: Distinct cycles—drop in 2021 and recovery later.

Seasonal: Captures regular fluctuations with short bursts of higher amplitude.

Residual: Captures volatility events, especially around 2020 – 2022.

4. USD/RUB

Trend: Highly volatile, reflecting war-related and sanctions-related economic shocks.

Seasonal: Spikes are extreme and localized; possibly artifacts from market disruptions.

Residual: Exceptionally large residuals (e.g., around 2022), signifying major structural shocks.

5. AUD/NZD

Trend: Fluctuating, smooth behavior with short cycles.

Seasonal: Consistent and stable seasonality, unlike the volatility seen in USD/RUB.

Residual: Well-behaved, clustered around zero.

STL Decomposition for Log Returns (Trend, Seasonal, Residual)

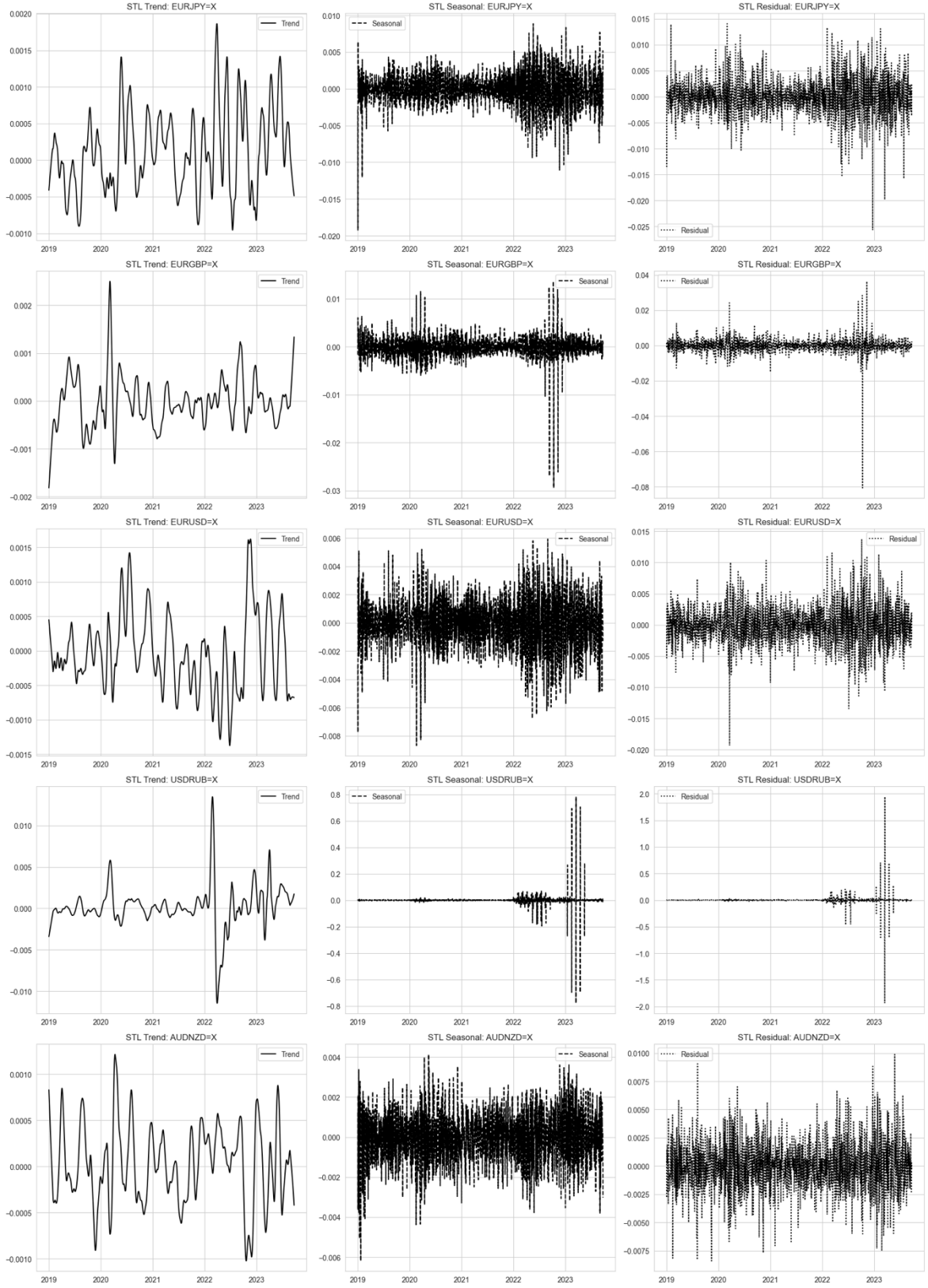


Fig. 13. STL decomposition for log returns of exchange rates

Figure 14 displays the Classical Decomposition of Squared Log Returns for various exchange rates.

1. EUR/JPY

Trend: Gradual increase in volatility from 2020 to 2022, then a slight decline.

Seasonal: Spikes in regular intervals suggest certain recurring volatility bursts.

Residual: Small and relatively stable, indicating a good fit from trend and seasonality.

2. EUR/GBP

Trend: Steady increase during 2020–2021, then a plateau at higher volatility levels.

Seasonal: Strong periodic spikes, possibly calendar or macroeconomic announcement effects.

Residual: Mostly low noise, but some residual spikes persist.

3. EUR/USD

Trend: Clear cyclical volatility pattern — peaks in 2020 and late 2022.

Seasonal: Regular seasonal bursts, less pronounced than in EUR/GBP.

Residual: Random noise, with some large outliers suggesting event-driven shocks.

4. USD/RUB

Trend: Huge jump in volatility in early 2022, possibly due to geopolitical tensions (e.g., Russia-Ukraine war).

Seasonal: Clear recurring seasonal spikes with high magnitude.

Residual: Appears sparse, with large spikes during crisis periods, but otherwise minimal.

5. AUD/NZD

Trend: Volatility increases consistently after 2021.

Seasonal: Frequent, but small seasonal fluctuations.

Residual: Low residual variance, implying a well-captured decomposition.

Classical Decomposition for Squared Log Returns (Trend, Seasonal, Residual)

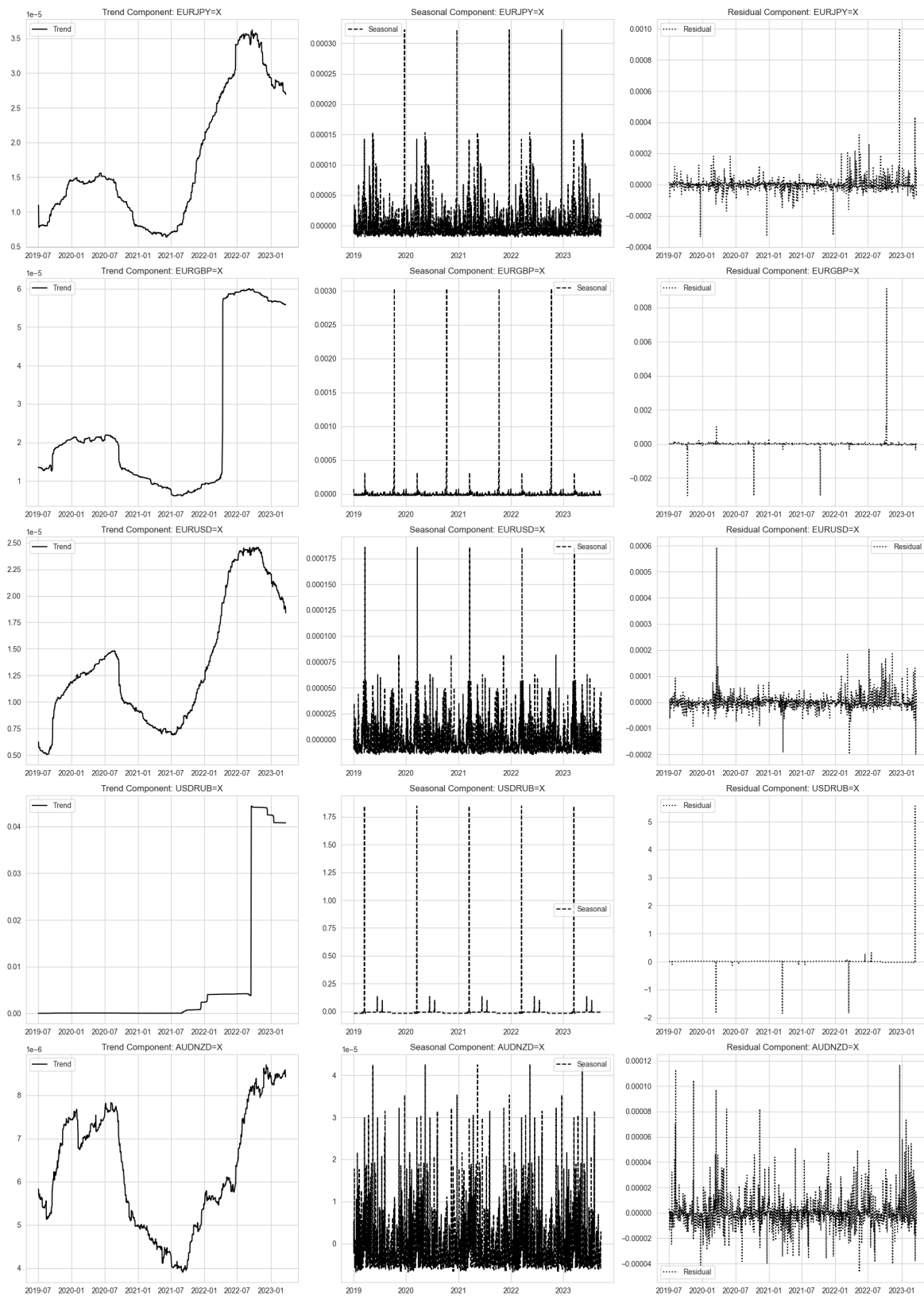


Fig. 14. Classical decomposition for squared log returns of exchange rates

Figure 15 presents the Seasonal-Trend decomposition using LOESS (STL) of the squared log returns for five major exchange rates: EURJPY, EURGBP, EURUSD, USDRUB, and AUDNZD.

EURJPY=X: The volatility trend exhibits multiple moderate spikes, especially during late 2021 and early 2022, aligning with periods of monetary policy tightening. The seasonal component shows minor cyclicalities, while the residuals remain centered around zero with moderate dispersion.

EURGBP=X: Volatility remains relatively low until 2022, after which an extreme spike is observed — likely reflecting financial market disruptions due to geopolitical tensions or macroeconomic shocks. Seasonal effects are minimal, while residuals capture the sudden outlier in volatility during this event.

EURUSD=X: This pair displays considerable shifts in volatility, with notable spikes during the COVID-19 outbreak in 2020 and the Fed tightening cycle in 2022. The trend component confirms rising volatility in recent years, while the seasonal and residual components capture short-term fluctuations.

USDRUB=X: The most extreme volatility among the five pairs is observed here, particularly during early 2022, coinciding with the onset of the Russia–Ukraine conflict. The trend component shows a dramatic spike, and the seasonal element reflects persistent, high-frequency fluctuations. The residuals further emphasize the intensity and irregularity of this volatility shock.

AUDNZD=X: Volatility remains relatively subdued, with cyclical fluctuations in the trend component. Seasonal patterns are more pronounced here compared to the other series, suggesting some regular intra-annual variation, likely due to synchronized macroeconomic activity between Australia and New Zealand.

STL Decomposition for Squared Log Returns (Trend, Seasonal, Residual)

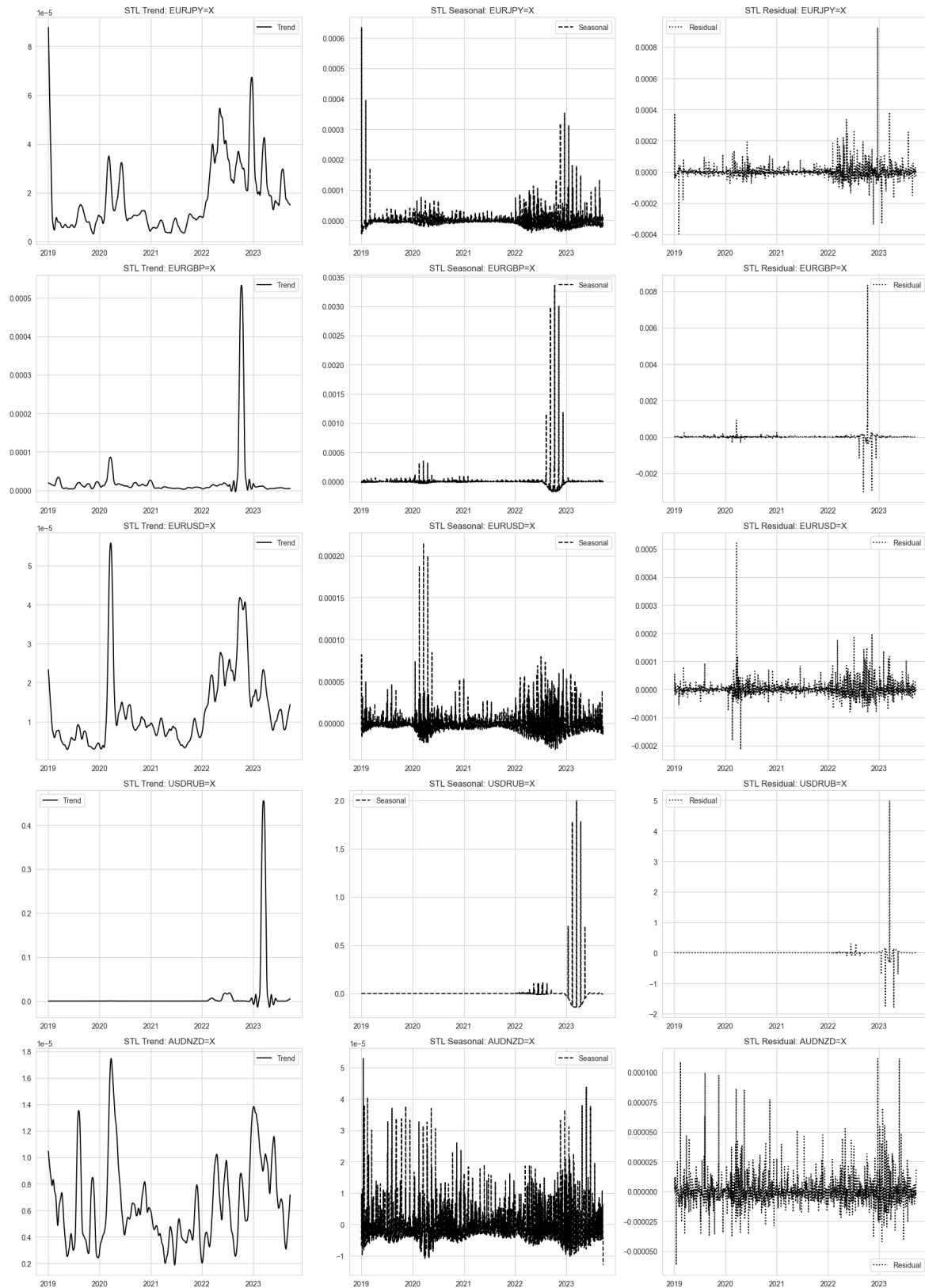


Fig. 15. STL decomposition for squared log returns of exchange rates

3.3 Modeling Implementation

In this section, we employ the Autoregressive Integrated Moving Average (ARIMA) model to forecast exchange rate series under three different transformations: price levels, log returns, and squared log returns (as a volatility proxy). ARIMA is a standard tool in time series forecasting, particularly well-suited to capturing autocorrelation structures in stationary and differenced time series.

The selection of ARIMA orders (p , d , q) was guided by the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of each series. For the price-level series, the orders were chosen to ensure stationarity either by differencing (when needed) or by selecting an autoregressive process with sufficient lags. For return series, which are typically stationary, most models were specified with $d=0$, and the values of p and q were based on statistically significant spikes in ACF/PACF plots.

The ARIMA orders used were as follows:

Price Levels:

EURUSD (2,0,0), EURGBP (2,0,0), EURJPY (2,1,0), USDRUB (3,0,0), AUDNZD (2,0,0)

Log Returns:

EURUSD (1,0,1), EURGBP (1,0,1), EURJPY (1,0,1), USDRUB (6,0,1), AUDNZD (10,0,0)

Squared Log Returns:

EURUSD (1,0,1), EURGBP (1,0,1), EURJPY (1,0,0), USDRUB (1,0,1), AUDNZD (2,0,0)

The figure 16 presents the forecast of the EUR/USD exchange rate price levels

To obtain a robust estimate of out-of-sample performance, we implemented time series cross-validation using a 10-fold TimeSeriesSplit approach. In each fold, the ARIMA model was re-estimated recursively for one-step-ahead forecasting, with the training set expanding at each step and the forecast produced dynamically.

The plot distinguishes between:

Actual values (shown in black): representing observed EUR/USD price levels.

Cross-validated forecasts (shown in blue): generated through the rolling-origin strategy across validation folds.

This recursive updating mechanism closely mimics a real-time forecasting scenario, where models are re-estimated as new data become available.



Fig. 16. ARIMA forecast for price levels of EUR/USD

The chart 17 illustrates the forecasted EUR/GBP exchange rate using an ARIMA(2,0,0) model, applied to the raw price level series.



Fig. 17. ARIMA forecast for price levels of EUR/GBP

While the model tracks the general level and short-term fluctuations reasonably well, it exhibits limitations during periods of heightened volatility — particularly in late 2022 and early 2023, where abrupt changes and spikes (e.g., possibly Brexit-related or policy shocks) lead to significant forecast deviations. This illustrates the well-known shortcoming of univariate linear models in capturing structural breaks and regime shifts.

The chart 18 illustrates the actual and forecasted values of the EUR/JPY exchange rate over a multi-year period.



Fig. 18. ARIMA forecast for price levels of EUR/JPY

The ARIMA model captures general trends and cyclical behaviors in the exchange rate data, although it may smooth out some of the more volatile spikes.

From 2019 to mid-2020, the model reasonably tracks the downward trend and subsequent mild recovery.

During the 2021–2023 period, the forecast aligns well with the upward momentum, reflecting the pair's consistent appreciation.

The 19th chart illustrates the historical and forecasted price levels of the USD/RUB exchange rate using an ARIMA model, with validation done via cross-validation.

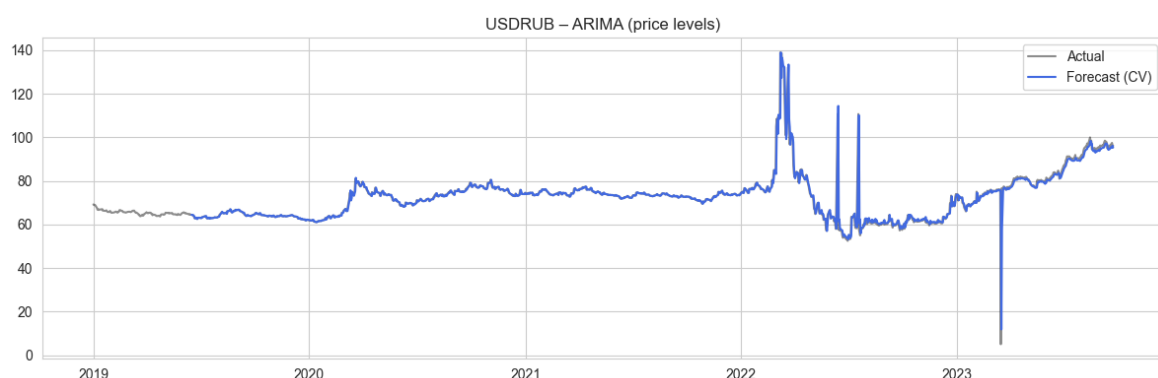


Fig. 19. ARIMA forecast for price levels of USD/RUB

From 2019 to early 2022, the ARIMA model captures the broad, relatively stable behavior of the USD/RUB pair.

However, significant spikes and volatility are observed starting in early 2022—coinciding with geopolitical tensions and economic sanctions — that caused the ruble to swing sharply.

The ARIMA model struggles in these periods of extreme market disruption, leading to erratic or unrealistic forecast values, including sharp outliers and large dips (e.g., near-zero values), which are not plausible.

The graph 20 illustrates actual and forecasted exchange rate values for the AUD/NZD pair using an ARIMA model with cross-validation.



Fig. 20. ARIMA forecast for price levels of AUD/NZD

The AUD/NZD exchange rate displays mean-reverting and oscillatory behavior, with movements largely bounded between 1.03 and 1.14.

The ARIMA model tracks these cyclical patterns relatively well, especially during periods of moderate volatility.

Short-term deviations and noise are evident but do not drastically affect the overall trend estimation.

21th chart presents an ARIMA forecast of EUR/USD log returns, a more stationary form of financial time series.

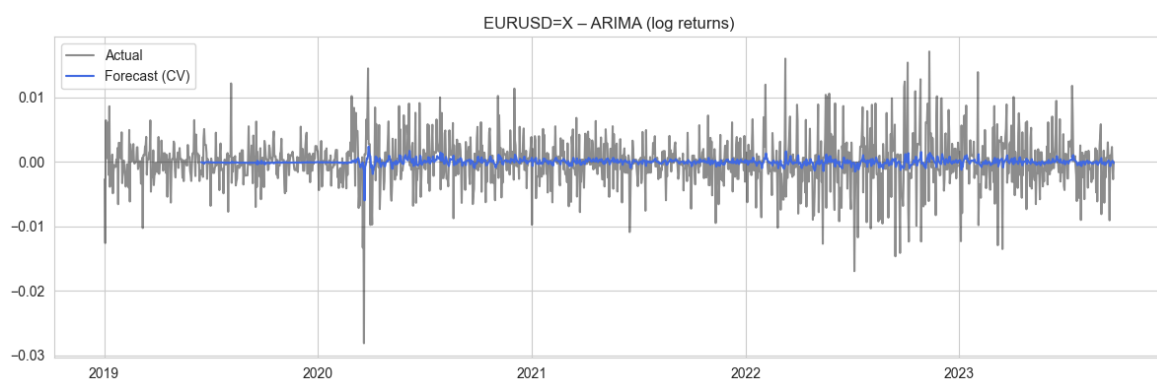


Fig. 21. ARIMA forecast for log returns of EURUSD

Log return data exhibit high volatility and randomness, which is typical for high-frequency financial returns.

The ARIMA model forecasts (blue line) tend to regress toward the mean (close to zero), indicating limited ability to predict directionality or magnitude of short-term returns.

Despite that, the model manages to maintain a stable forecast without extreme deviation, which aligns with ARIMA's design assumptions.

The chart 22 shows an ARIMA-based forecast of EUR/GBP log returns, which are commonly used for modeling exchange rate dynamics due to their stationarity and interpretability.

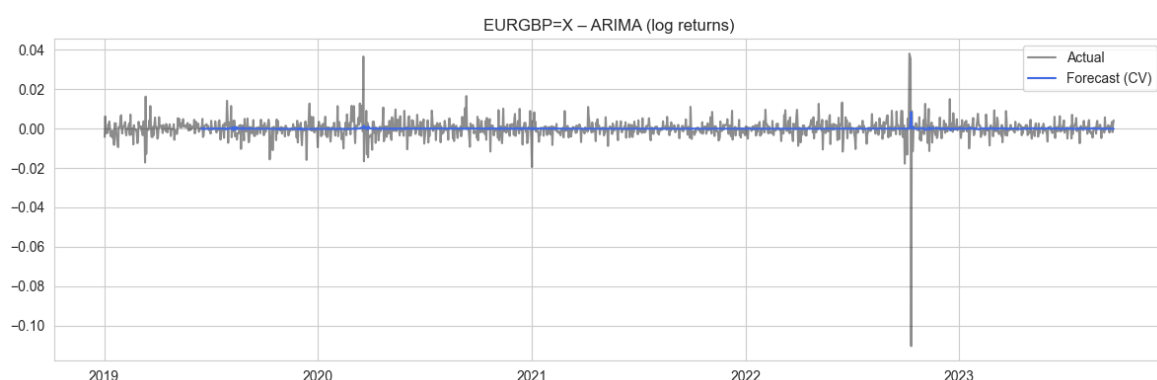


Fig. 22. ARIMA forecast for log returns of EUR/GBP

The log returns are largely centered around zero, indicating a lack of strong trend.

The ARIMA model's forecasts (blue line) closely follow the mean, showing limited deviation, and generally underpredicting extreme values.

One notable outlier in late 2022 shows a dramatic negative spike in actual returns that ARIMA fails to anticipate — highlighting the model's limitations in capturing rare, high-impact events.

Chart 23 visualizes the ARIMA model's forecast of EUR/JPY log returns, which offers a deeper look at the short-term dynamics of exchange rate changes.

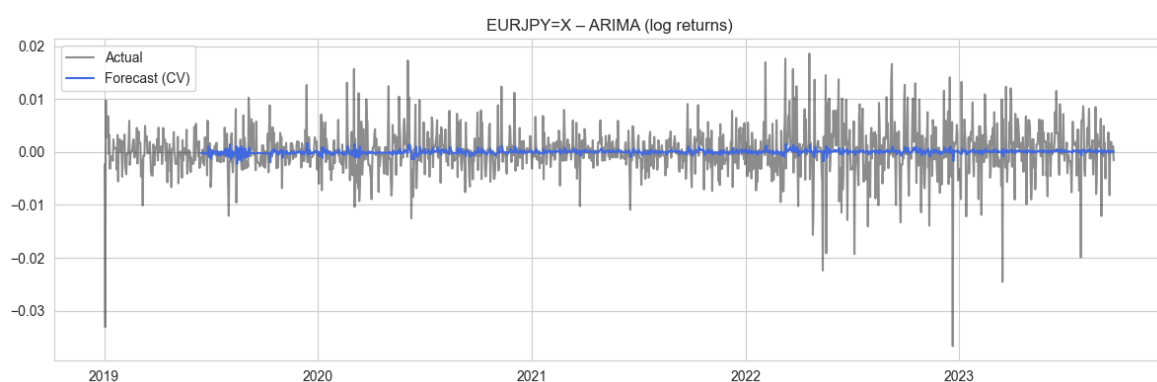


Fig. 23. ARIMA forecast for log returns of EUR/JPY

The ARIMA model's forecast (blue) is relatively smooth and hovers around the zero line, suggesting a mean-reverting tendency.

The model performs well during stable periods but struggles to anticipate volatility spikes, especially noticeable in early 2023 and late 2022.

The actual return series shows fat tails and volatility clustering, which are not captured by the ARIMA model.

The figure 24 presents the ARIMA-based forecast of the USDRUB exchange rate, where the model is applied to the log return series.

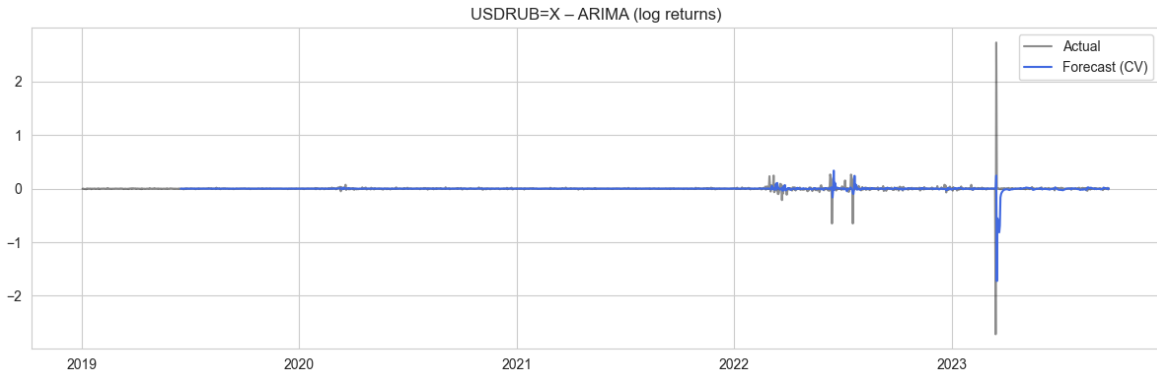


Fig. 24. ARIMA forecast for log returns of USD/RUB

Notably, the ARIMA model struggles to adapt to abrupt regime shifts and extreme events, such as the 2022 Russian invasion of Ukraine and the resulting market turbulence. This is reflected in its failure to capture large spikes and crashes, where actual returns far exceed the model's forecast range. Despite this, ARIMA provides reasonable forecasts during more tranquil periods, when return dynamics follow a more linear and mean-reverting behavior.

This figure 25 illustrates the performance of an ARIMA model applied to the log returns of the AUDNZD exchange rate.

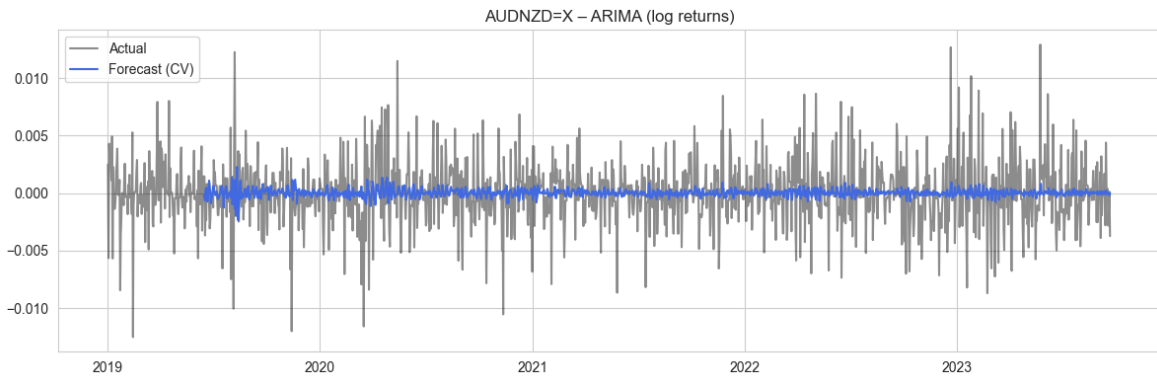


Fig. 25. ARIMA forecast for log returns of AUD/NZD

The ARIMA model captures the conditional mean of the return process but not the conditional variance, making it poorly suited to periods of high volatility. This is evident in its inability to reflect the spikes and troughs observed in the actual data, particularly during risk-on/risk-off episodes driven by macroeconomic shifts or central bank policy divergences between Australia and New Zealand.

This figure 26 displays the ARIMA model's forecast performance for the squared log returns of the EURUSD exchange rate.

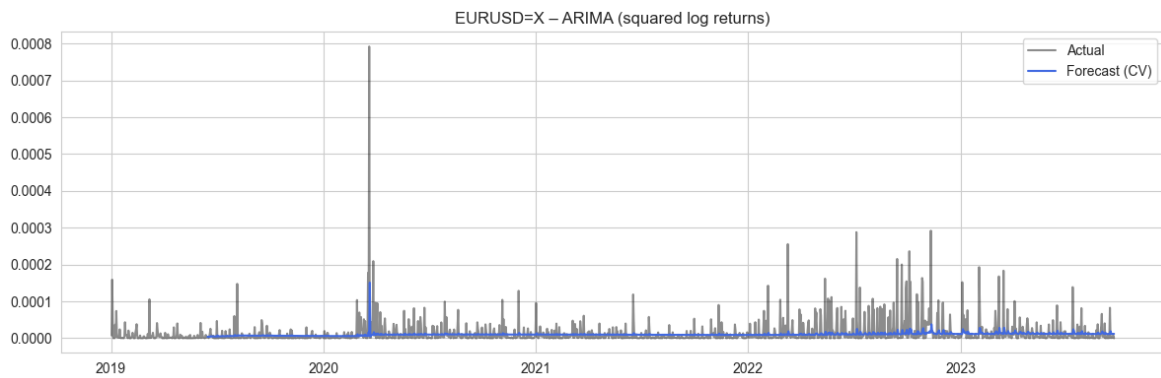


Fig. 26. ARIMA forecast for squared log returns of EUR/USD

The ARIMA model, designed for mean forecasting, struggles to adequately model time-varying volatility. While it provides a smoothed baseline of expected variance, it fails to capture volatility clustering, a key feature of financial return series. This limitation is especially visible during episodes of high market stress, where actual squared returns surge and forecasts lag behind.

The figure 27 presents the performance of an ARIMA model in forecasting the squared log returns of the EUR/GBP exchange rate, which serve as a proxy for the currency pair's volatility.

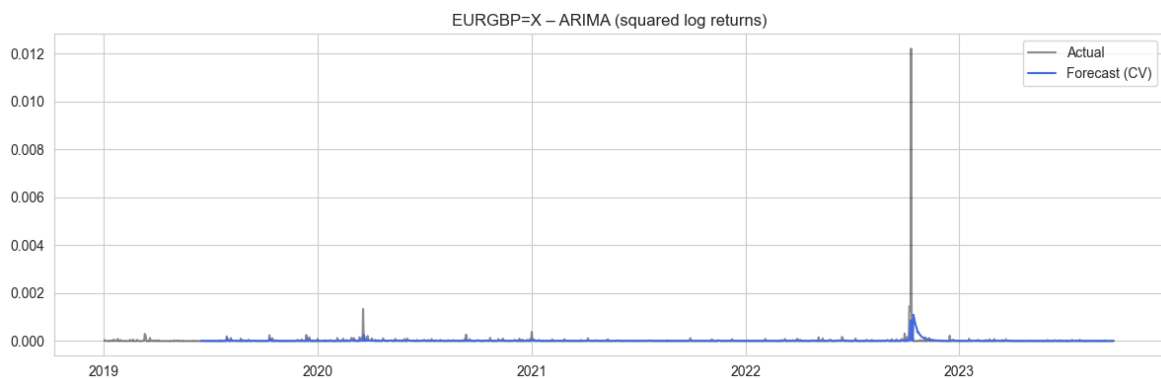


Fig. 27. ARIMA forecast for squared log returns of EUR/GBP

From the plot, it is evident that the ARIMA model captures the low and moderate fluctuations in volatility reasonably well throughout most of the period from 2019 to 2023. However, it fails to anticipate the large volatility spike observed in late 2022, as reflected by a sharp and isolated spike in the actual series that significantly exceeds the model's forecast. This discrepancy suggests that while ARIMA models are capable of modeling steady-state behavior, they may struggle with extreme events or structural breaks, which are common in financial time series due to macroeconomic shocks or geopolitical events.

This figure 28 illustrates the forecast performance of an ARIMA model on the squared log returns of the EUR/JPY exchange rate.

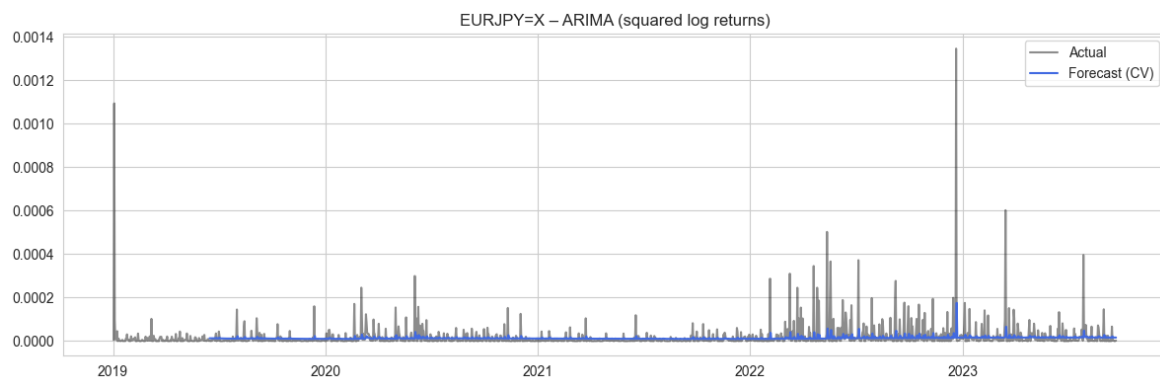


Fig. 28. ARIMA forecast for squared log returns of EUR/JPY

Throughout the time span from 2019 to 2023, the ARIMA model appears to consistently underestimate the magnitude of volatility, especially during periods of market turbulence. Notable volatility spikes in early 2019 and late 2022 to early 2023 are poorly captured by the forecast, with the predicted values remaining considerably flat and failing to adapt to these abrupt increases.

The chart 29 presents the actual versus forecasted squared log returns for the USD/RUB exchange rate using an ARIMA model.

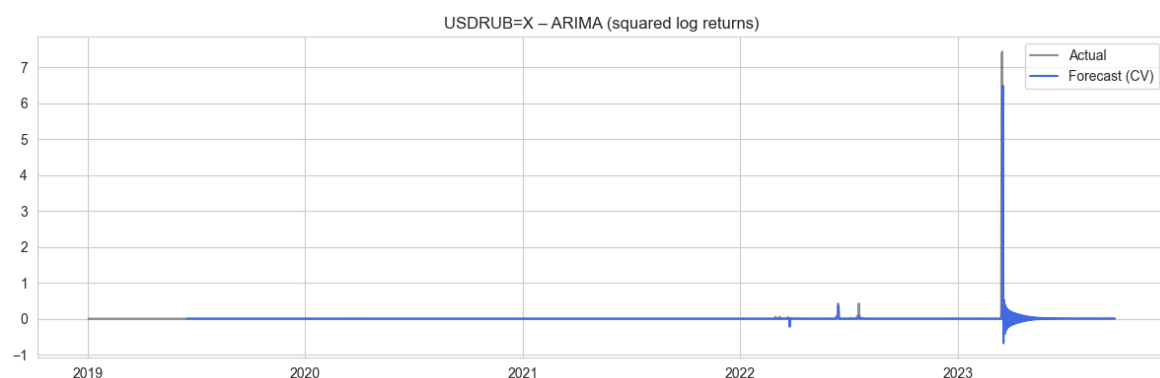


Fig. 29. ARIMA forecast for squared log returns for USD/RUB

From 2019 through mid-2022, the time series of squared returns appears mostly stable and near-zero, indicating minimal volatility. However, the model begins to diverge significantly in 2022, with the forecast producing implausibly

extreme values, including negative squared returns (which are not mathematically valid), particularly around early 2023.

This instability in the forecast indicates model breakdown in the presence of sudden volatility shocks, likely corresponding to geopolitical disruptions or market crises, which are common in emerging markets like Russia. The ARIMA model clearly fails to cope with these dynamics, producing erratic and unbounded forecasts that do not reflect the actual volatility structure of the data.

Plot 30 compares the actual and forecasted squared log returns for the AUD/NZD exchange rate using an ARIMA model.

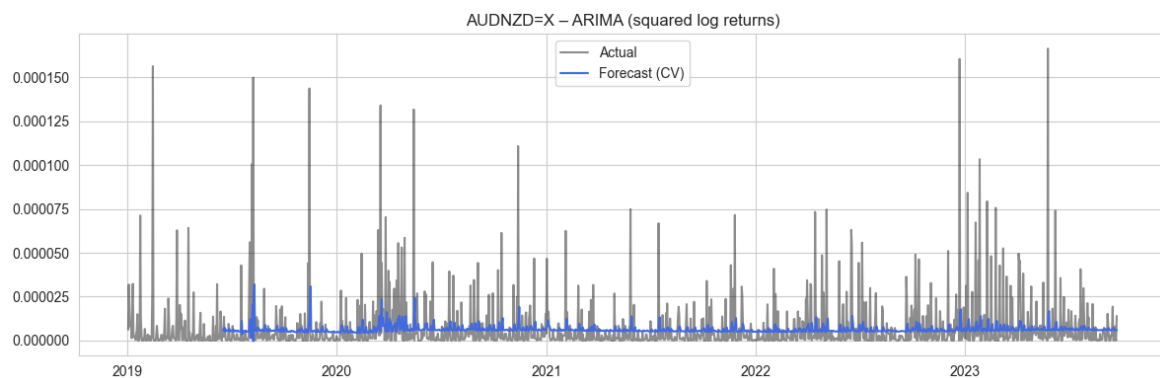


Fig. 30. ARIMA forecast for squared log returns of AUDNZD

Throughout the time period from 2019 to early 2023, the actual series displays frequent and abrupt volatility spikes, reflecting the high-frequency fluctuations and occasional jumps typical of financial return series. In contrast, the ARIMA forecasts remain consistently smooth and low in magnitude, failing to capture the intensity and variability of the volatility shocks.

Although ARIMA can track the general low-variance trend, it systematically underestimates the magnitude of volatility during turbulent periods. This mismatch is due to ARIMA's limitation in modeling conditional heteroskedasticity, which is a key feature of financial return series.

The graph 31 presents the LSTM model's performance in forecasting EUR/USD exchange rate price levels over time. The black line represents the actual historical exchange rate, while the blue line shows the predicted values from the LSTM model on the test (cross-validation) set, which consists of the final 20% of the data.

The model was trained with the following configuration:

- Window size: 10 previous observations used to predict the next one.
- LSTM units: 256 neurons to capture temporal dependencies.
- Epochs: 30 training iterations.
- Batch size: 16.
- Scaling: The series was standardized using StandardScaler before training and then inverse-transformed for comparison.



Fig. 31. LSTM forecast for price levels of EUR/USD

The predicted values follow the overall shape of the actual price, suggesting that the LSTM captured short-term trends and reversals reasonably well. The fact that the forecast starts where the actual line ends confirms that the model was only evaluated out-of-sample (not on the training set), as is consistent with good modeling practice.

This graph 32 shows the LSTM model's prediction of the EUR/GBP exchange rate in price levels.



Fig. 32. LSTM forecast for price levels of EUR/GBP

The forecast (blue line) begins around late 2022, which marks the start of the test period (final 20%).

The model captures the overall range and fluctuations of the actual data, including the mean-reverting behavior of EUR/GBP.

A sharp spike right before the forecast zone (late 2022) suggests market volatility or a rare event, which may have challenged the model slightly in the transition to the test period.

Despite this, the forecast stabilizes quickly and follows the actual price path reasonably well, indicating the LSTM's robustness to sudden shifts when enough historical context is provided.

This figure 33 illustrates the forecasting performance of a Long Short-Term Memory (LSTM) neural network applied to the EURJPY exchange rate in its raw price level form.



Fig. 33. LSTM forecast for price levels of EUR/JPY

The model appears to capture the general upward trend in EURJPY during 2022–2023, demonstrating that the LSTM can effectively learn and replicate long-term directional movements from historical data. Notably, the model maintains good alignment during stable periods and even manages to approximate turning points, although with some visible lag or attenuation in volatility.

Despite being a non-linear, data-driven model, the LSTM shows smooth and consistent forecasts without severe overfitting or erratic behavior, which speaks to the robustness of the model's design and hyperparameter tuning. However,

it slightly underpredicts the peaks in early 2023, likely due to the model's tendency to average out extreme values.

This figure 34 presents the performance of a Long Short-Term Memory (LSTM) model trained to forecast the USDRUB exchange rate using raw price level data.



Fig. 34. LSTM forecast for price levels of USD/RUB

The USDRUB series exhibits extreme volatility and structural breaks, especially surrounding the 2022 geopolitical shock, with sharp spikes and regime shifts. Despite these challenges, the LSTM model generally captures the broad direction of the price movements in the post-2022 period. The forecasts track the upward drift in 2023 fairly well, maintaining alignment with the general trend.

However, there is a notable forecasting error around early 2023, where the model generates a large, spurious drop in the predicted exchange rate. This likely reflects the difficulty LSTMs face when encountering sudden regime shifts or out-of-distribution events, particularly when those events are rare or unseen during training. The error could also stem from data sparsity or interpolation artifacts around the war-induced market dislocations.

This figure 35 illustrates the performance of an LSTM model trained to predict the AUDNZD exchange rate using daily price levels.



Fig. 35. LSTM forecast for price levels of AUD/NZD

The model exhibits solid performance, particularly in capturing the turning points and directional movements of the exchange rate. Following the strong upward trend in 2022, the LSTM successfully anticipates the broad downward correction in late 2022, as well as the subsequent range-bound fluctuations in 2023.

The forecasts generally track the actual series closely, suggesting that the LSTM is able to learn and generalize from the underlying temporal patterns in the data. The relatively low volatility and mean-reverting nature of AUDNZD compared to more geopolitically sensitive currencies (e.g., USDRUB) likely contributed to the LSTM's stability and accuracy in this case.

The figure 36 demonstrates the predictive performance of the LSTM model on EUR/USD daily log returns.



Fig. 36. LSTM forecast for log returns of EUR/USD

Unlike the ARIMA model, which assumes linearity and stationarity, the LSTM shows better adaptation to changes in volatility regimes.

However, it still lags during sharp return fluctuations, particularly during high-volatility periods (e.g., 2022–2023).

The chart 37 shows the EUR/GBP log return forecasts using an LSTM (Long Short-Term Memory) model, and compares it against actual market returns.

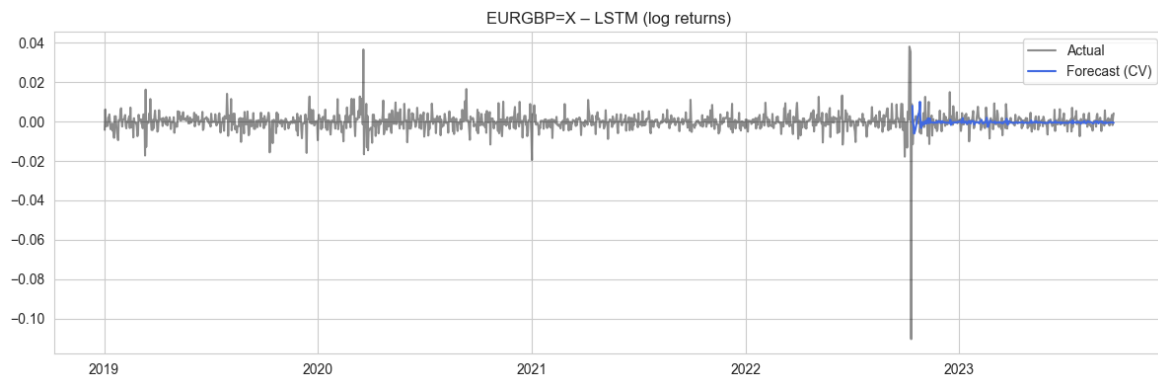


Fig. 37. LSTM forecast for log returns of EUR/GBP

The LSTM predictions (blue) start from late 2022, likely due to the data split for training vs. validation.

The forecasts remain centered near zero, consistent with the general nature of daily FX returns, which tend to be mean-reverting with low daily volatility.

The model misses sharp spikes, such as the extreme drop seen in late 2022, indicating limited ability to predict rare or abrupt market events.

The chart 38 shows the EUR/JPY log return forecasts using an LSTM (Long Short-Term Memory) model

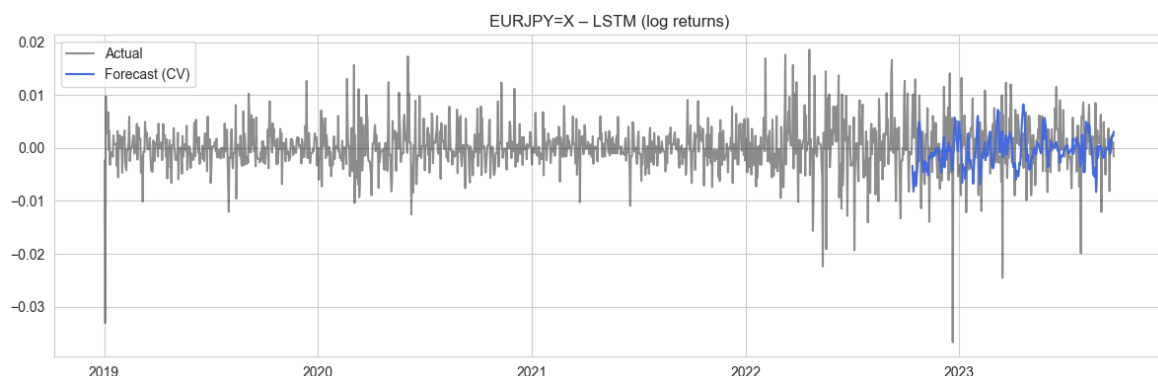


Fig. 38. LSTM forecast for log returns of EUR/JPY

The forecasted values are tightly clustered around zero—typical of daily FX log returns, which are usually low in magnitude and mean-reverting.

The model captures general noise and trends in periods of moderate activity. Misses sharp movements or extreme returns, such as volatility spikes and abrupt drops (e.g., early 2023).

The figure 39 demonstrates the predictive performance of the LSTM model on USD/RUB daily log returns.

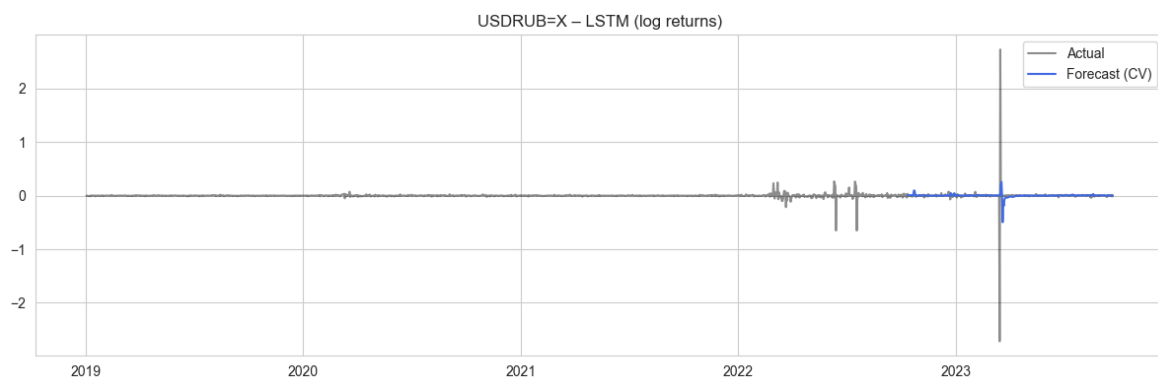


Fig. 39. LSTM forecast for log returns of USD/RUB

Unlike previous currency pairs, this chart shows very large spikes (especially a massive negative return around early 2023), which are orders of magnitude larger than usual log returns.

This suggests a major market shock, likely due to geopolitical or macroeconomic events affecting the Russian Ruble.

The LSTM forecast (blue) remains tightly clustered near zero, completely missing the magnitude and occurrence of these outliers.

This is a significant limitation of the model — it was likely trained on mostly stable periods and couldn't generalize to such rare, extreme cases.

The figure 40 demonstrates the predictive performance of the LSTM model on USD/RUB daily log returns.

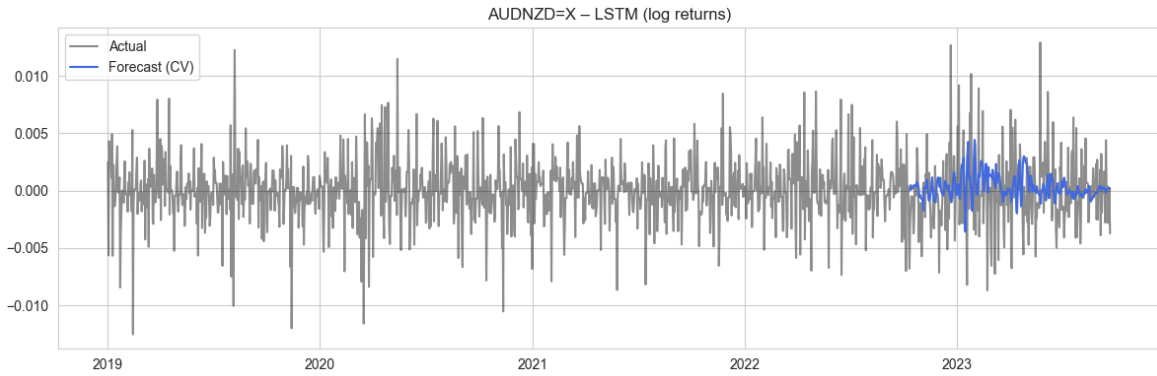


Fig. 40. LSTM forecast for log returns of AUD/NZD

The forecast follows the mean-reverting nature of log returns, staying centered near zero.

Compared to the actuals, the model is somewhat responsive to volatility changes —forecasted variation slightly increases during periods of heightened volatility (early 2023), though it's still smoother than the real data.

The model misses sharp jumps and underestimates the amplitude of extreme moves.

However, it captures the general direction and shape of returns better than in more volatile pairs like USDRUB.

This plot 41 shows the squared log returns for the EUR/USD exchange rate using an LSTM model.

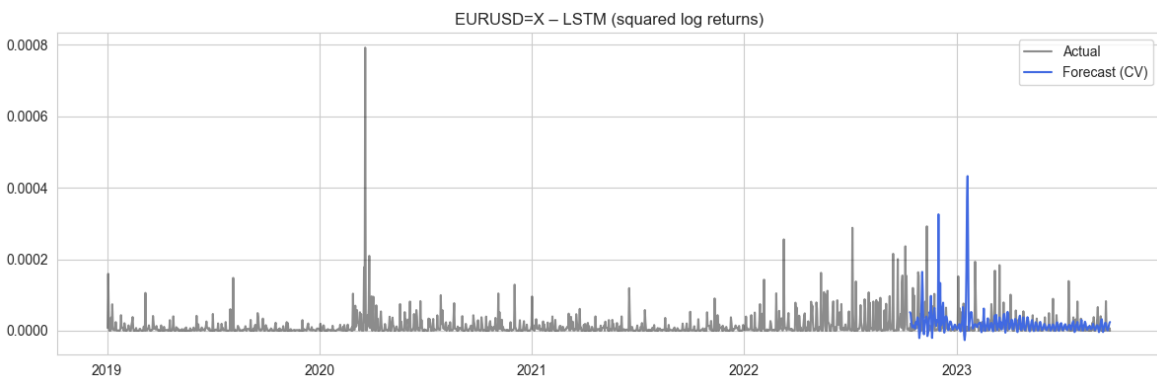


Fig. 41. LSTM forecast for squared log returns of EUR/USD

High spikes in actual returns are visible around early 2020 (likely related to COVID-19 market shocks) and late 2022 to early 2023.

Forecast accuracy appears relatively good post-2022, especially in terms of capturing the general volatility pattern, though the magnitude of sharp spikes is

often underestimated — a common trait of LSTM models trained on squared returns.

This figure 42 presents the results of an LSTM model trained to forecast the volatility of the EURGBP exchange rate, with volatility proxied by squared log returns.

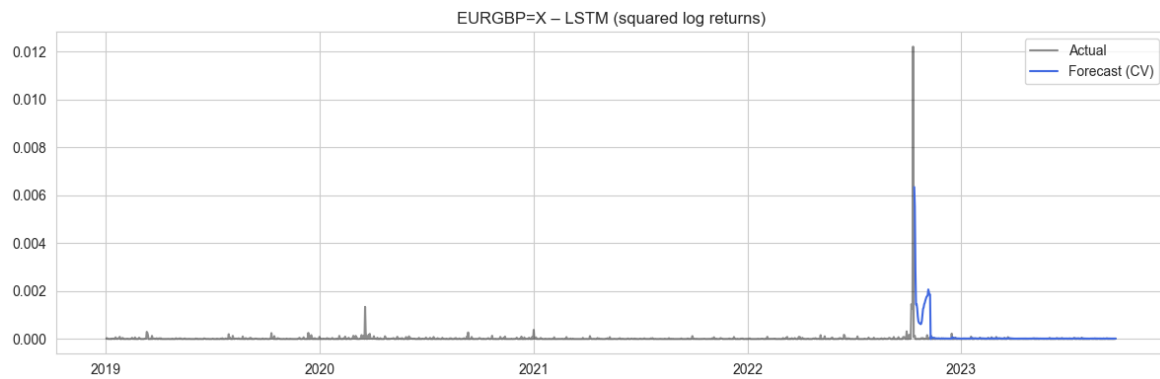


Fig. 42. LSTM forecast for squared log returns of EUR/GBP

The LSTM captures the general clustering behavior of volatility, including the large spike in 2022, which likely reflects a period of significant market turbulence (e.g., political or macroeconomic shocks affecting EUR or GBP). This major event appears to have been partially learned by the model, as it forecasts an upward volatility shift afterward, although it underestimates the peak magnitude.

Following this spike, the LSTM continues to reflect a gradual reversion to the mean in volatility, consistent with financial time series properties like volatility clustering and mean reversion. However, the forecast path tends to be smoother than the actual data, which often includes sharp, transient spikes.

This figure 43 illustrates the volatility forecasting performance of an LSTM model applied to the EURJPY exchange rate.

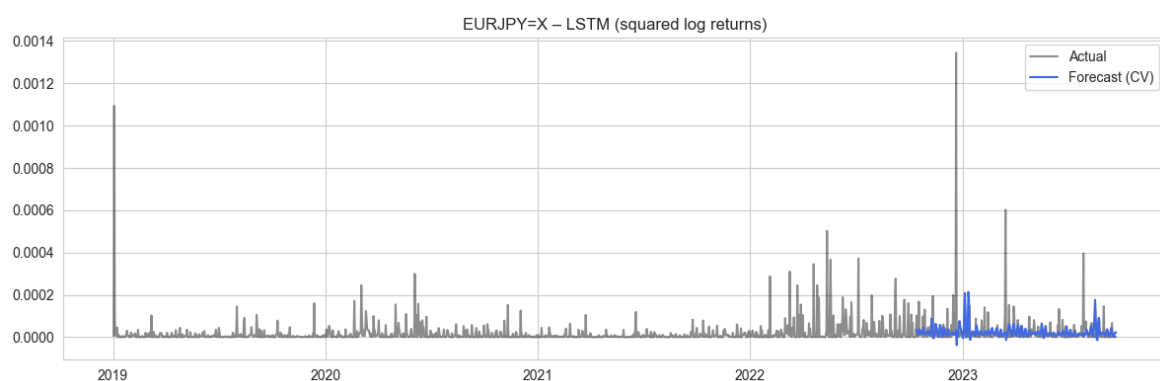


Fig. 43. LSTM forecast for squared log returns of EUR/JPY

The LSTM model reasonably captures low-to-moderate volatility regimes, particularly from late 2022 into 2023. The forecast reacts to shifts in volatility, indicating that the model has learned short-term temporal dependencies. Notably, however, the model struggles with forecasting sudden large spikes, especially the high-volatility shock in early 2023. This underprediction is common when models are trained only on historical price dynamics without incorporating external shock indicators or event-driven features.

The graph 44 shows the LSTM model's prediction of volatility (measured as squared log returns) for the USD/RUB exchange rate.



Fig. 44. LSTM forecast for squared log returns of USD/RUB

The squared log returns are near-zero for most of the timeline, except: a dramatic volatility spike around early 2022, which corresponds to major geopolitical events (e.g. Russia-Ukraine conflict, sanctions, capital controls).

The LSTM fails to capture this major spike. The forecast (blue line) remains low and smooth throughout the test period.

After the spike, the volatility quickly reverts to low levels, and the LSTM continues forecasting minimal volatility.

The graph 45 displays the forecasting of volatility (measured as squared log returns) for the AUD/NZD exchange rate using an LSTM model.

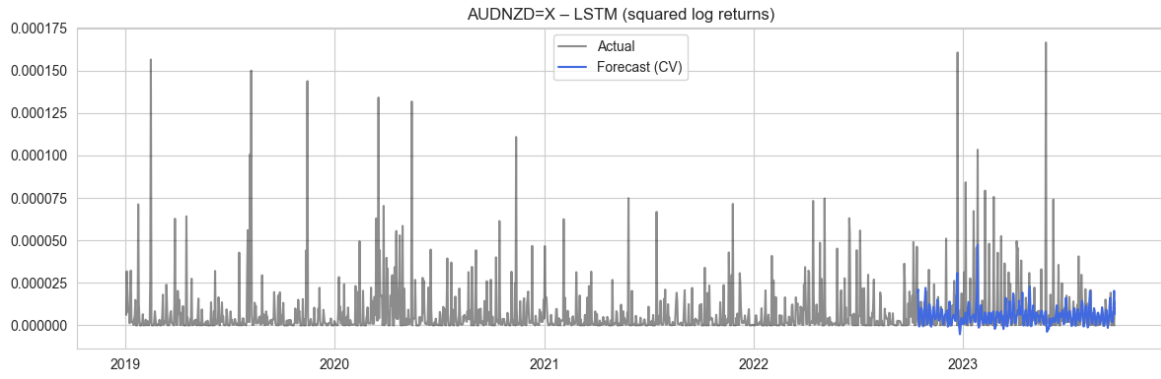


Fig. 45. LSTM forecast for squared log returns of AUD/NZD

The actual squared returns (black line) show a frequent pattern of small values, punctuated by intermittent volatility spikes.

These spikes are not extreme (in contrast to USDRUB), but regular and noisy.

The LSTM prediction (blue line) captures the baseline level of volatility well, remaining close to the general trend of the actual data. However, the model smooths out the peaks, especially the taller volatility spikes. For instance, noticeable spikes in early 2023 are not fully captured in the forecast.

The forecast often lags behind sudden jumps, a common limitation of purely historical time series models.

3.4. Evaluation and Comparison

To assess the performance of the forecasting models, we compare the Mean Squared Error (MSE) across different currency pairs and target transformations (price levels, log returns, and squared log returns). The two models under comparison are ARIMA, a classical linear time series model, and LSTM (Long Short-Term Memory), a recurrent neural network designed to capture nonlinear dependencies and temporal patterns.

Currency Pair	ARIMA (MSE)	LSTM (MSE)
EURUSD	1.627×10^{-5}	2.137×10^{-5}
EURGBP	1.947×10^{-5}	1.998×10^{-5}
EURJPY	3.318×10^{-1}	2.231
USDRUB	1.418×10^1	2.454×10^1
AUDNZD	7.419×10^{-6}	1.033×10^{-5}

EURUSD and EURGBP: Both models performed comparably, with ARIMA slightly outperforming LSTM. This suggests that for stable and liquid currency pairs, linear models can perform well even without complex feature learning.

EURJPY and USDRUB: LSTM produced significantly higher errors compared to ARIMA, especially for USDRUB where market conditions were more volatile and structurally non-stationary. This underperformance may stem from LSTM's sensitivity to regime shifts or extreme events not well captured in training data.

AUDNZD: Both models showed low MSE, though ARIMA had a marginal edge. This again reflects the effectiveness of simpler models when the series is less volatile and follows stable trends.

Currency Pair	ARIMA (MSE)	LSTM (MSE)
EURUSD=X	1.350×10^{-5}	1.882×10^{-5}
EURGBP=X	2.442×10^{-5}	1.144×10^{-5}
EURJPY=X	1.811×10^{-5}	3.472×10^{-5}
USDRUB=X	1.319×10^{-2}	4.394×10^{-2}
AUDNZD=X	6.600×10^{-6}	9.510×10^{-6}

EURGBP=X was the only series where LSTM outperformed ARIMA, showing its ability to potentially capture subtle nonlinearities in medium-frequency return movements.

For the other pairs, ARIMA consistently produced lower MSEs, particularly in the case of USDRUB=X, where LSTM exhibited large errors. This may be due to the high volatility and possible regime changes in the Russian ruble during the period, which neural networks trained on historical patterns might struggle to generalize.

Currency Pair	ARIMA (MSE)	LSTM (MSE)
EURUSD=X	1.216×10^{-9}	2.496×10^{-9}
EURGBP=X	8.999×10^{-8}	3.582×10^{-7}
EURJPY=X	2.939×10^{-9}	7.952×10^{-9}
USDRUB=X	9.508×10^{-2}	3.191×10^{-1}
AUDNZD=X	2.251×10^{-10}	3.039×10^{-10}

ARIMA outperformed LSTM across all pairs, sometimes by wide margins (e.g., USDRUB=X).

The LSTM's higher error suggests difficulties in capturing the heteroskedastic nature of financial time series, which ARIMA, although not designed for volatility modeling per se, handled better in these cases.

To statistically evaluate the forecasting performance differences, Diebold–Mariano (DM) and Giacomini–White (GW) tests were performed. The tests compare predictive accuracy across models using forecast error differentials.

The results of both DM and GW tests strongly favor ARIMA across all currency pairs at conventional significance levels ($p < 0.01$), with all test statistics positive and significant. This suggests that ARIMA forecasts are statistically more accurate than those from LSTM for price levels.

Pair	DM Statistic (p)	GW Statistic (p)
EURUSD	4.06 (0.000)	6.90 (0.000)
EURGBP	6.55 (0.000)	10.75 (0.000)
EURJPY	15.03 (0.000)	25.81 (0.000)
USDRUB	7.18 (0.000)	10.92 (0.000)
AUDNZD	7.93 (0.000)	13.38 (0.000)

For log returns, the majority of test results are negative and statistically significant (DM and GW), indicating that LSTM outperforms ARIMA for these series. Notably, EURJPY=X and AUDNZD=X show the strongest evidence of LSTM superiority. However, for the USDRUB=X pair, the performance difference was not statistically significant.

Pair	DM Statistic (p)	GW Statistic (p)
EURUSD=X	-2.20 (0.028)	-2.61 (0.009)
EURGBP=X	-1.88 (0.060)	-2.04 (0.041)
EURJPY=X	-3.63 (0.000)	-4.40 (0.000)
USDRUB=X	-0.87 (0.386)	-0.87 (0.386)
AUDNZD=X	-2.33 (0.020)	-2.69 (0.007)

When evaluating squared log returns (a proxy for volatility forecasts), the statistical tests do not show consistent superiority for either model. For most

series, the test statistics were insignificant, suggesting that neither model dominates. A marginal advantage for ARIMA is suggested by EURGBP=X and EURUSD=X in GW tests.

Pair	DM Statistic (p)	GW Statistic (p)
EURUSD=X	-1.12 (0.262)	-1.35 (0.176)
EURGBP=X	-1.60 (0.110)	-1.96 (0.049)
EURJPY=X	-0.50 (0.617)	-0.51 (0.607)
USDRUB=X	-1.14 (0.253)	-1.31 (0.189)
AUDNZD=X	1.75 (0.081)	1.73 (0.083)

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